Inflation Inequality Across the Euro Area[1]

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Abstract

We document inflation inequality in the euro area during the recent energy crisis using harmonized HICP prices and household budget surveys. We construct household-level and group indices and counterfactual price paths excluding household price policies. In 2022, the annual-average individual P20–P80 range reaches 4.8 percentage points, compared with a 1.2-point gap between mean inflation in the first and fifth income quintiles. Differences between household groups therefore capture only part of inflation inequality. Fiscal policies designed to alter consumer prices can redistribute measured inflation exposure across fixed household baskets. Counterfactual indices indicate that tariff shields, VAT cuts, fuel rebates, and price caps reduced annual-average inflation by 1.7 points, the individual P20–P80 range by 1.8 points, and the Q1–Q5 gap by 0.6 point in 2022.

Keywords: Inflation, Distribution of Income, Fiscal Policy
JEL codes: E31, D30, C43

[1]This paper presents the authors' views and should not be interpreted as reflecting the views of the Banque de France or the Eurosystem. This paper primarily uses HBS household microdata, supplemented by published tabulations from Eurostat. The responsibility for all conclusions drawn from the data lies entirely with the authors. We thank Pradana Aumars and Lyna El Kamel for excellent research assistance. We thank Michael Ehrmann for helpful discussions.

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Introduction

This paper documents how the recent inflation surge affected euro-area households. Inflation rates in the euro area rose sharply from mid-2021, displaying considerable dispersion across countries and households. The 2021–2023 episode was a large relative-price shock, initially driven by imported energy and commodity prices.

Inflation inequality emerges when relative-price movements interact with heterogeneous household expenditure shares. If all prices rose proportionally, fixed-basket inflation would be identical across households. We examine how recent energy-price shocks and the fiscal response shaped household inflation exposure across the euro area. Specifically, we construct monthly price indices at the household level and for household groups defined by income, age, household size, and area of residence in each euro-area country, following Eurostat's HICP methodology. We use individual 2020 HBS expenditure records both to construct household-specific indices and to aggregate expenditure into household-group baskets. The household-level indices show dispersion within groups as well as differences between group averages. We also construct counterfactual price indices that remove the main household price policies of 2021–2023. This counterfactual is a direct consumer-price accounting exercise: it isolates the contribution of these policies to household inflation, holding expenditure weights fixed between scenarios and abstracting from behavioral responses. Throughout the paper, a policy effect is defined as observed minus counterfactual inflation.

Our main result is that household-level inflation dispersion rises sharply during the shock and is substantially wider than differences between household groups. In 2022, the annual individual P20–P80 range reaches 4.8 percentage points, while the annual Q1–Q5 gap reaches 1.2 points. A model saturated with all interactions among income, age, household size, and residence explains only 9.1% of within-country variance on average across months, ranging from 5.1% to 13.7%. The remaining 90.9% lies within observationally similar household groups. Income quintiles thus capture one systematic dimension of heterogeneity rather than the full distribution of household outcomes.

The Q1–Q5 inflation gap nevertheless widens during the shock and varies across euro-area countries. In 2022, it exceeds 3 points in Estonia, Latvia, Lithuania, and the Netherlands and reaches 2.9 points in Ireland. Older and rural households also experience higher inflation during the rise in energy prices. Household price policies reduce annual-average inflation by 1.7 percentage points, the individual P20–P80 range by 1.8 points, and the Q1–Q5 gap by 0.6 point in 2022.

The literature measures household inflation by combining price changes with differences in expenditure patterns. For the United States, Hobijn and Lagakos (2005) reweight aggregate price indices for demographic groups. More recently, Jaravel (2024) constructs long-run distributional consumer price indices for the United States, closely following the BLS methodology. Using household-level expenditure or scanner data, Argente and Lee (2021), Kaplan and Schulhofer-Wohl (2017) document household-level dispersion within broad income and demographic groups. Using barcode-level grocery purchases in the United States, O'Flaherty (2026a) shows that demographic group averages mask most household-level inflation dispersion. In the euro area, Ampudia et al. (2024) show how household shopping behavior shapes the effect of monetary policy on inflation heterogeneity along the income distribution. Pallotti et al. (2024) combine consumption, income, and balance-sheet data to measure welfare losses across euro-area households. Building on Charalampakis et al. (2022), we take a complementary approach to O'Flaherty (2026a). Depending on the exercise, we cover 18 to 20 euro-area countries and the full consumption basket, assigning a common country-product HICP series to harmonized HBS expenditure categories. The policy counterfactual first reconstructs component price paths at finer detail before mapping them to this common basket support. This broader scope permits harmonized cross-country comparisons and monthly counterfactuals for household price policies. Our RAS-calibrated baskets also aggregate to official HICP item weights. The trade-off is that our measures capture heterogeneity in expenditure shares, but not differences in prices paid, product quality, or substitution within product categories (Sangani, 2026).

A related literature evaluates the fiscal response to the recent inflation surge. Using household microsimulation models, Pallotti et al. (2024) and Amores et al. (2025) show that fiscal interventions reduced welfare losses and inequality, although broad price policies were less targeted than income support. Using detailed UK household data, Levell et al. (2026) highlight a trade-off between targeting and efficiency: energy price subsidies better protect households with high exposure to energy-price shocks, including within income groups, but distort consumption choices, while transfers preserve price signals but may be less well targeted. For Germany, Krebs and Weber (2024) examine conditions under which energy price controls can be optimal, while Dertwinkel-Kalt and Wey (2025) show that the design of the energy price brakes can induce suppliers to raise prices while reinforcing households' incentives to save energy. Bonnet et al. (2025) show, for France, that income-targeted transfers are more effective than fuel price subsidies at limiting welfare losses relative to household income. We instead combine harmonized household inflation measures with monthly counterfactual price paths reconstructed at the COICOP-component level for all 20 euro-area countries. This approach identifies which products drove the distributional effect of

household price policies: lower electricity and gas prices reduce the Quintile 1–Quintile 5 inflation gap, while fuel policies partly offset that reduction because higher-income households spend more on private transport.

The remainder of the paper is organized as follows. Section 1 describes the data and methodology. Section 2 examines developments in inflation inequality in the euro area. Section 3 presents counterfactual estimates of the heterogeneous effects of household price policies. Section 4 turns to the inflation burden.

1 Data and Methodology

Our approach has three core steps. We match HICP product-price indices to HBS expenditure records, calibrate the resulting household and household-group baskets to official HICP product weights, and aggregate the corresponding price indices across countries. This section focuses on those choices and on the statistical objects used in the results. The appendices provide the full RAS algorithm, national data exceptions, alternative income definitions, rent conventions, and validation tests.

1.1 Sources, Samples, and Product Coverage

Sources and Samples

The analysis combines public HICP data with restricted Household Budget Survey (HBS) microdata and national HBS inputs. Headline HICP inflation measures price changes for the aggregate household consumption basket. Each national HICP uses price changes for goods and services weighted by household expenditure from national accounts, household budget surveys, and other national sources. Eurostat then aggregates the national indices with country weights that reflect each country's share of euro-area household consumption expenditure.[1]

Eurostat updates the HICP product weights annually. The new weights enter with the January release and remain fixed within the calendar year. Let $\mathcal{J}_y^{\text{HICP}}$ denote the set of product categories with official HICP weights in year y, $p_{j,y,m}$ the price index for product j in month m, and $p_{j,y,0}$ its price reference for the weight year, corresponding to the previous December. The official

[1]The calculation follows Regulation (EU) 2016/792 and common data-collection practices. HICP products follow the European Classification of Individual Consumption according to Purpose (ECOICOP), and historical series are mapped to the classification used in the HBS. National statistical institutes compile the HICP in collaboration with Eurostat.

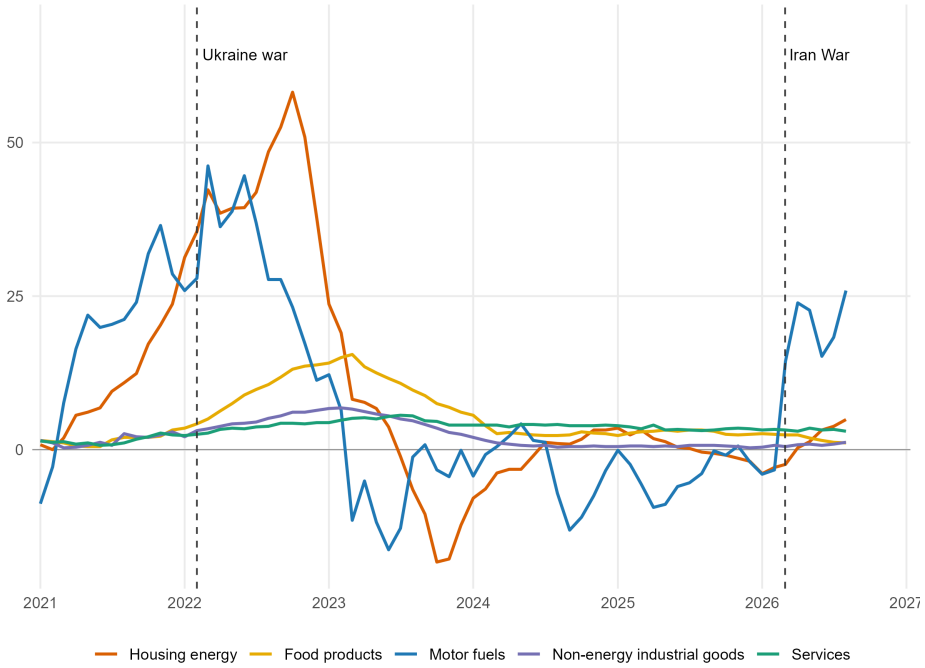
within-year Laspeyres link and chained index are

$$P_{y,m} = \sum_{j \in \mathcal{J}_y^{\text{HICP}}} \omega_{j,y}^{\text{HICP}} \frac{p_{j,y,m}}{p_{j,y,0}}, \tag{1}$$

$$I_{y,m} = I_{y-1,12} P_{y,m}. \tag{2}$$

Figure 1 shows inflation across the main HICP components in the euro area since January 2021. Energy and fuel prices rise sharply from mid-2021 and peak in 2022. Inflation subsequently declines across most components, with energy prices falling more sharply than food prices and other components.

Figure 1: Inflation developments in the euro area (year on year, in %)



Sources: Eurostat HICP euro-area aggregate; authors' calculations.

Our primary expenditure source is the 2020 HBS microdata. The records contain expenditure by product, total net household income, and household characteristics. Group-level indicators cover all 20 euro-area countries through the national and tabulation-based exceptions described below. Household distributions cover 178,268 households in 19 countries; only Portugal is excluded because its individual expenditure divisions cannot separate rents. Italy enters these household-level calculations without an income filter. These households remain fixed across price years, so the monthly series does not represent a panel of newly observed consumption choices. Figure 4

follows this sample from January 2021 to December 2025. Analyses that jointly use income, age, household size, and degree of urbanization require non-missing values for all four characteristics and therefore use a complete-case sample of 152,440 households.

The 2020 HBS wave combines surveys collected over country-specific reference periods. Fieldwork dates and survey frequency therefore need not coincide across countries. The fixed wave provides a cross-section of consumption heterogeneity rather than annual observations on changing household baskets.[2] Repeated Spanish HBS waves suggest that broad differences in expenditure shares between low- and high-income households remain relatively stable even when aggregate consumption patterns change (Appendix Figure A.1).

Product Coverage and Treatment of Rents

All HBS-based calculations in the main text use the three-digit ECOICOP1 classification and never use finer household-expenditure detail. This is the finest common household-level detail that retains France and Germany in the broad cross-country sample. Depending on the country, this level represents roughly 45–50 product groups in the source HBS; countries with finer expenditure codes are aggregated to this common support, and an unmatched three-digit component is represented by its nearest common parent. Source price series must be complete over the full calculation period before aggregation, which keeps the product composition within each three-digit category fixed over time. The policy module can reconstruct price interventions at four-digit detail, but these price paths are aggregated to the same three-digit support before they are combined with HBS baskets. Historical and validation exercises retain their explicitly stated matching levels. Appendix B compares the individual P20–P80 inflation range across two- to five-digit COICOP levels and reports the matching level used in each specification.

The baseline excludes actual and imputed rents and removes the corresponding HICP components before matching. Parent categories are then rebuilt, and the remaining HBS expenditure and HICP product weights are normalized over the common non-rent support. This coverage applies to household indices, group indices, and policy counterfactuals. The resulting measures capture non-rent consumption-price exposure rather than an all-items cost-of-living index. Table C.1 reports alternative rent conventions, while Appendices E and F report the corresponding normalization and validation exercises.

[2] See Accardo et al. (2023) on estimating changes in consumption inequality between survey waves.

Household Groups and National Exceptions

We construct groups by income, age of the household reference person, area of residence, and household size. Residence groups use the Degree of Urbanisation (DEGURBA).[3] For the two-group residence comparisons, we combine towns and suburbs with rural areas and refer to this combined category as the rural group; cities form the other group.

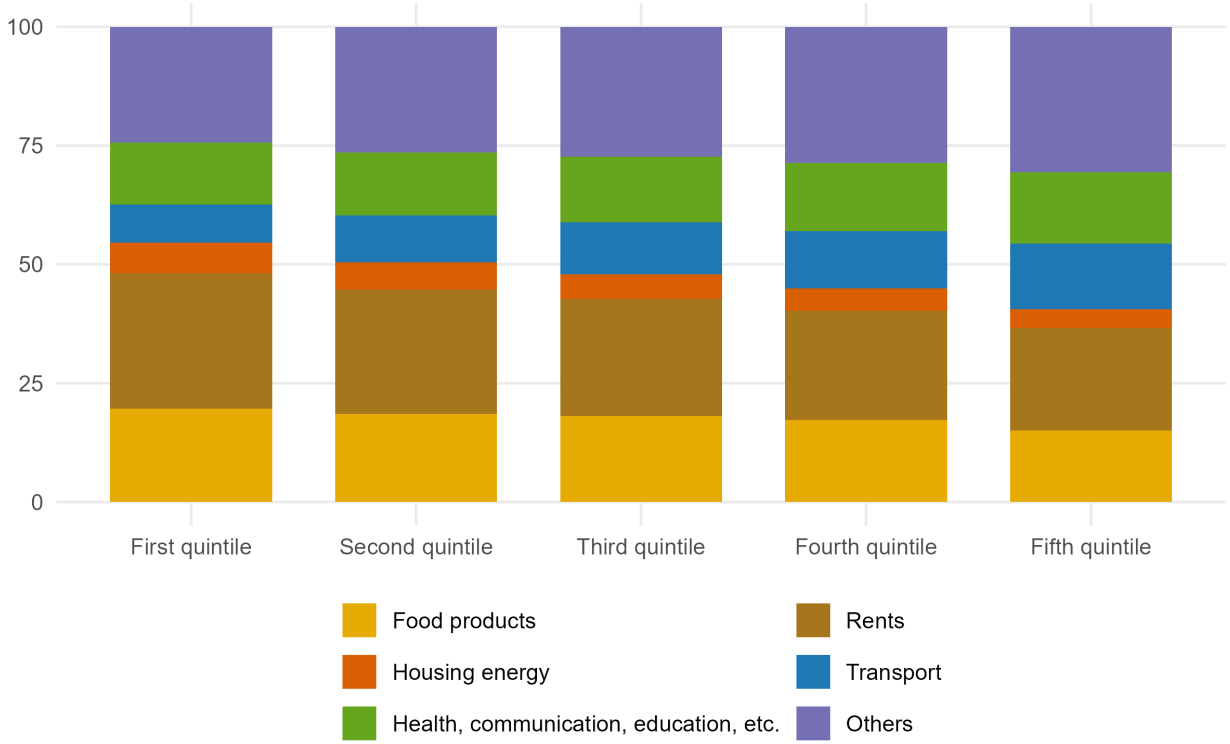
We use HBS survey weights throughout the construction of nationally representative group baskets. Income groups rank households by total net household income without adjustment for household size, so they describe household income rather than equivalized living standards. National income-quintile thresholds are defined separately within each national HBS sample and survey wave. Missing income excludes a household from income baskets but not from age, residence, or unconditional analyses when the required variables are available. Appendix C motivates this choice, relates it to Klick and Stockburger (2024), and compares it with equivalized income while holding household inflation rates, the sample, and weights fixed.

The main 2021–2023 analysis uses fixed 2020 HBS baskets. For each national group, we aggregate individual product expenditure into a representative consumption basket. Three national exceptions preserve broader group coverage. Italy uses the national group baskets described in Appendix G. Portugal uses published group tabulations for non-rent inflation because its individual expenditure divisions cannot isolate rents. Malta uses published residence-area baskets and relative-expenditure weighting because the required groups and RAS margins are unavailable.

Figure 2 compares income-quintile baskets at the euro area level. Lower-income households allocate larger expenditure shares to food and housing. Within housing, household energy is retained in the inflation indices and is central to the analysis. Exposure to a given price shock therefore depends on both these basket differences and the products affected by the shock. Housing energy and private-transport expenditure are also higher in the rural group than in cities, reflecting differences in heating needs and mobility constraints.

[3] DEGURBA uses a common European methodology based on a 1 km^2 population grid. It distinguishes cities (densely populated areas), towns and suburbs (intermediate-density areas), and rural areas (thinly populated areas). Urban centers have at least 50,000 inhabitants and 1,500 inhabitants per km^2; urban clusters have at least 5,000 inhabitants and 300 inhabitants per km^2; the remaining areas are rural.

Figure 2: Consumption baskets by income quintile in the euro area (in % of total expenditure)



Notes: Country budget shares are averaged using 2020 HICP country weights, normalized across the 20 countries and held common across quintiles. These are national income quintiles, not euro-area income ranks. Rents comprise actual and imputed rents; the inflation indices exclude both. Microdata supply the baskets except for the documented national and tabulation-based fallbacks.
Sources: HBS microdata, Eurostat, Istat, authors' calculations.

1.2 Calibration and Price Indices

RAS Calibration

Unadjusted HBS expenditure shares do not generally aggregate to the annual HICP product weights. We therefore apply Row and Column Adjustment Scaling (RAS) to the household–product expenditure matrix. The calibration preserves relative HBS expenditure profiles while matching two margins: each household's share of retained national expenditure and each product's normalized HICP weight. Normalizing the calibrated cells within each household yields household-specific product weights. For household groups, we apply the same procedure after replacing household margins with each group's share of total expenditure. In both cases, the expenditure-weighted

average of the calibrated baskets reproduces the official product shares on the matched non-rent support. Appendix E provides the full algorithm, notation, convergence rule.

Household-Level Price Indices

Let $p_{c,j,y,m}$ denote the product price index in month m of year y, and let $\mathcal{J}^{\mathrm{M}}_{c,y}$ denote the matched non-rent product set for country c. Product indices are first expressed relative to the preceding December, aggregated with annual household weights, and then chained to the preceding December's household index:[4]

$$P_{ic,y,m} = \sum_{j \in \mathcal{J}^{\mathrm{M}}_{c,y}} \omega_{j,ic,y} \frac{p_{c,j,y,m}}{p_{c,j,y-1,12}}, \qquad I_{ic,y,m} = I_{ic,y-1,12} P_{ic,y,m}. \tag{3}$$

The household application uses fixed 2020 expenditure profiles, while annual HICP margins update aggregate product composition.

Group-Level Price Indices

For each income, age, or residence group, we aggregate the HBS expenditure records before applying RAS. Group indices therefore need not equal averages of separately calibrated household indices. Suppressing country subscripts, the within-year link and chained group index are

$$P_{g,y,m} = \sum_{j \in \mathcal{J}^{\mathrm{M}}_{y}} \omega_{j,g,y} \frac{p_{j,y,m}}{p_{j,y-1,12}}, \qquad I_{g,y,m} = I_{g,y-1,12} P_{g,y,m}.$$

Here the country subscript is suppressed, so $\mathcal{J}^{\mathrm{M}}_{y}$ is the matched non-rent product set. The baseline uses the 2020 HBS wave throughout; historical exercises using successive waves retain their stated matching convention.

1.3 National and Euro-Area Aggregation

We use two aggregation procedures for two distinct objects. First, the distributional bands in Figure 4 summarize national household-inflation distributions. Within each country, we calculate the mean and the 10th, 20th, 80th, and 90th empirical percentiles of household year-on-year inflation. We then average each national statistic using Eurostat's annual HICP country expenditure weights. For a national statistic $s_{c,y,m}$, the reported series is

$$s_{y,m} = \sum_{c} \omega_{c,y} s_{c,y,m}, \qquad \omega_{c,y} = \frac{W_{c,y}}{\sum_{k \in \mathcal{C}} W_{k,y}},$$

[4] See Eurostat's `hicp` R package, which implements the Eurostat HICP aggregation.

where $W_{c,y}$ is the published all-items HICP country weight and $\mathcal{C}$ is the fixed country sample. These bands average national percentile endpoints; they are not percentiles of a pooled euro-area distribution. The country weights are not additionally adjusted for rents.

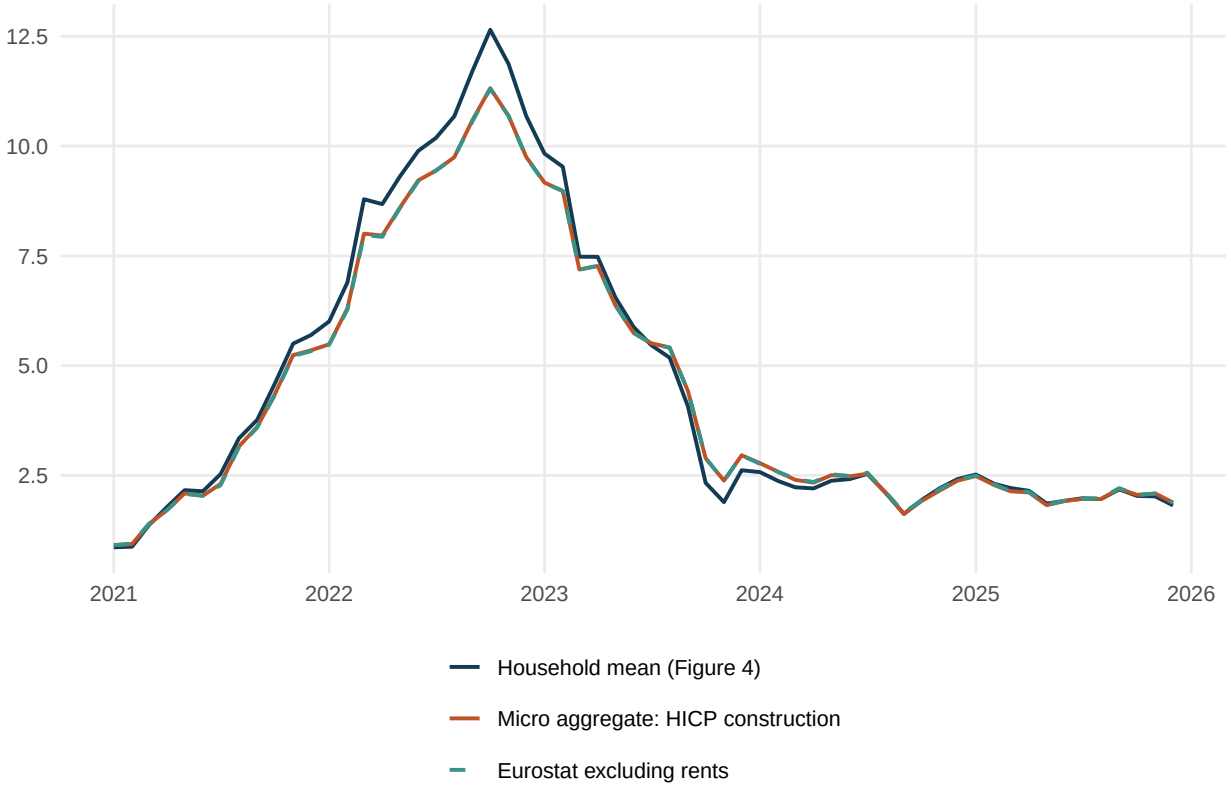
Second, aggregate household and group price indices require aggregation of December-relative price links before chaining. Let $q_{c,y}$ denote the annual HICP country expenditure weight after removing rent expenditure and normalizing over the included countries, so $\sum_c q_{c,y} = 1$. The household aggregate is

$$P_{c,y,m} = \frac{\sum_i a_{ic} E_{ic,y} P_{ic,y,m}}{\sum_i a_{ic} E_{ic,y}}, \tag{4}$$

$$P_{y,m} = \sum_c q_{c,y} P_{c,y,m}, \qquad I_{y,m} = I_{y-1,12} P_{y,m}. \tag{5}$$

The same sequence applies to group indices using group-specific expenditure masses. Averaging household inflation rates directly does not implement this annual chain aggregation. All baseline inflation results exclude actual and imputed rents. Official headline series and the explicitly identified rent-sensitivity and validation exercises retain their stated comparison coverage. Figure 3 reconciles the household mean and the chained micro aggregate with the matching Eurostat non-rent benchmark for the same 19-country sample.

Figure 3: Micro–macro reconciliation: 19-country sample excluding rents (year on year, in %)



Notes: All series cover the same 19 countries; Portugal is excluded. The household mean first constructs each household's chained index as in equation (3), calculates year-on-year inflation, and then averages the national means. The micro aggregate instead averages the household December-relative links using survey-weighted matched expenditure, applies identical annual non-rent country expenditure weights, and chains the resulting aggregate links as in equation (3). The analogous unnumbered expression for pre-aggregated household-group baskets is not used directly in this household-level reconciliation. The household mean is the Figure 4 series.
Sources: HBS microdata, Eurostat HICP, authors' calculations.

2 Main Results

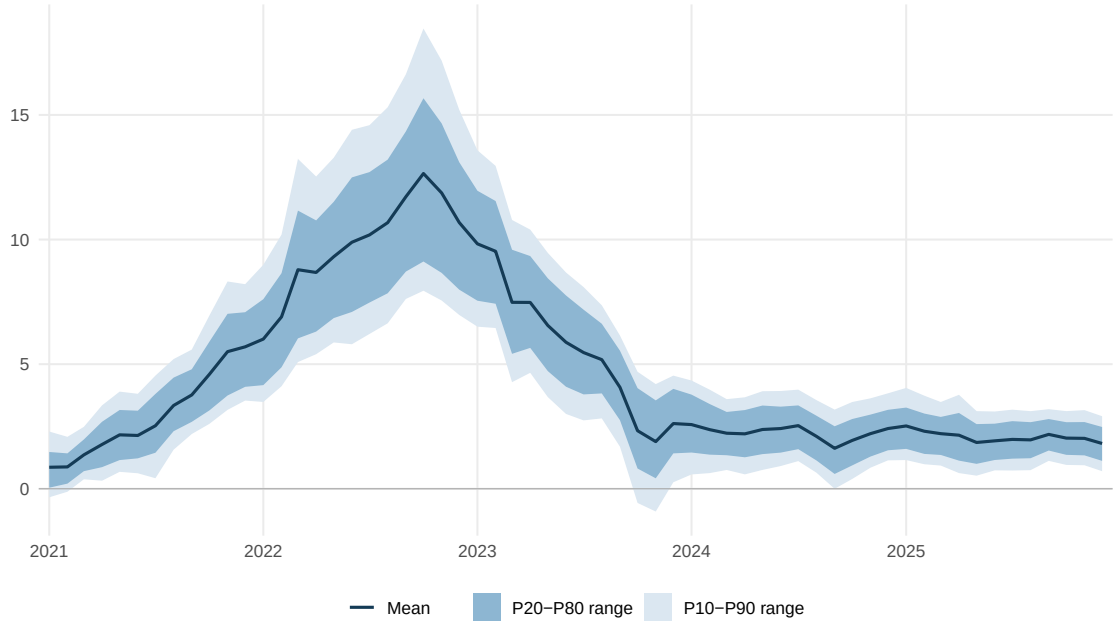
This section first establishes the magnitude of individual household inflation dispersion using the P20–P80 range, then asks how much of that dispersion observable characteristics jointly explain. It next turns from this variance accounting to economically interpretable shifts in distributions and group means across income, age, residence, and household-size groups. The final two subsections decompose individual dispersion and the income-group inflation gap by product and document how both measures differ across countries during the 2021–2023 energy crisis.

2.1 Household-Level Inflation Dispersion

The individual P20–P80 range of annual household inflation reaches 4.8 percentage points in 2022. Figure 4 shows how this dispersion develops month by month alongside country-weighted national means and the wider P10–P90 range from January 2021 to December 2025. The monthly P20–P80 range peaks at 6.5 points in October 2022, when mean inflation reaches 12.6%. In that month, the weighted national P10 and P90 cutoffs are 7.9% and 18.5%, respectively. The inflation surge therefore coincides with greater dispersion across household baskets within countries.

Although the underlying household sample is fixed, the households located at each national cutoff can change from month to month. The bands therefore describe the cross-sectional distribution in each month. Moreover, unlike the transaction data used by Kaplan and Schulhofer-Wohl (2017), our data do not capture differences in prices paid by households for the same product within a country.

Figure 4: Household inflation: country-weighted national mean, P20–P80 and P10–P90 ranges, 2021–2025 (year on year, in %)

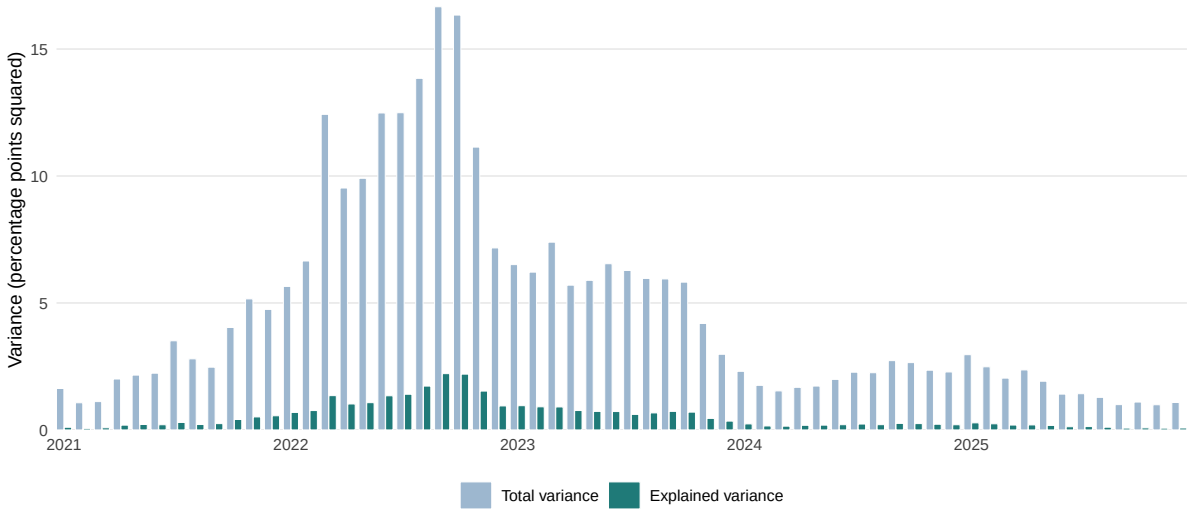


Notes: Monthly year-on-year inflation for 178,268 households in 19 euro-area countries, using fixed 2020 HBS microdata baskets. Portugal is excluded because the required household-product detail is unavailable; Italy enters without an income filter. Actual and imputed rents are excluded. Household inflation is constructed at the three-digit ECOICOP1 level. Household indices use annual RAS calibration with expenditure margins at that level. National means and distributional cutoffs first use HBS survey weights, without an income filter. Each national statistic is then averaged using annual Eurostat HICP country weights normalized over these 19 countries. The solid line shows the mean; the darker and lighter bands show averages of national P20–P80 and P10–P90 ranges. These are neither pooled euro-area ranges nor confidence intervals.
Sources: Eurostat HICP, HBS 2020 microdata, authors' calculations.

Household inflation dispersion moves closely with mean inflation. We next ask how much of this dispersion is associated with observable household characteristics. For each country and month, we estimate a saturated model that interacts national income quintile, age of the household reference person, household size, and degree of urbanization.

Figure 5 shows, for each month, total household inflation variance (light bars) and the part associated with the saturated demographic cells (dark bars). Dispersion rises sharply during the inflation surge and then recedes: total variance peaks at 16.7 percentage points squared in September 2022, while explained variance peaks at 2.2. The dark bar remains small relative to the light bar throughout the period. Observable characteristics account for 9.1% of total variance on average, with monthly shares between 5.1% and 13.7%. In separate one-way classifications, income explains 2.5% of variance on average, age 1.8%, urbanization 1.6%, and household size 1.5%. These shares are not additive because each estimate comes from a separate partition of households. The dark bars instead use fully interacted demographic cells and therefore also capture the explanatory power of combinations of characteristics. Even at the peak of household inflation dispersion, most variance thus remains within groups that share the same observed characteristics.

Figure 5: Household inflation: total and explained variance by month, 2021–2025



Notes: Grouped bars report monthly total variance, V, and explained variance, R^2V, in percentage points squared. Separate unweighted saturated models are estimated for each country and month using 152,440 households in 18 countries. Household inflation is constructed at the three-digit ECOICOP1 level. Cells are defined by the full interaction of national income quintile, age of the household reference person, household size, and degree of urbanization. No product variables enter the model. National R^2 values and within-country variances are aggregated using annual HICP country weights normalized over the sample.

Sources: Eurostat HICP, HBS 2020 microdata, authors' calculations.

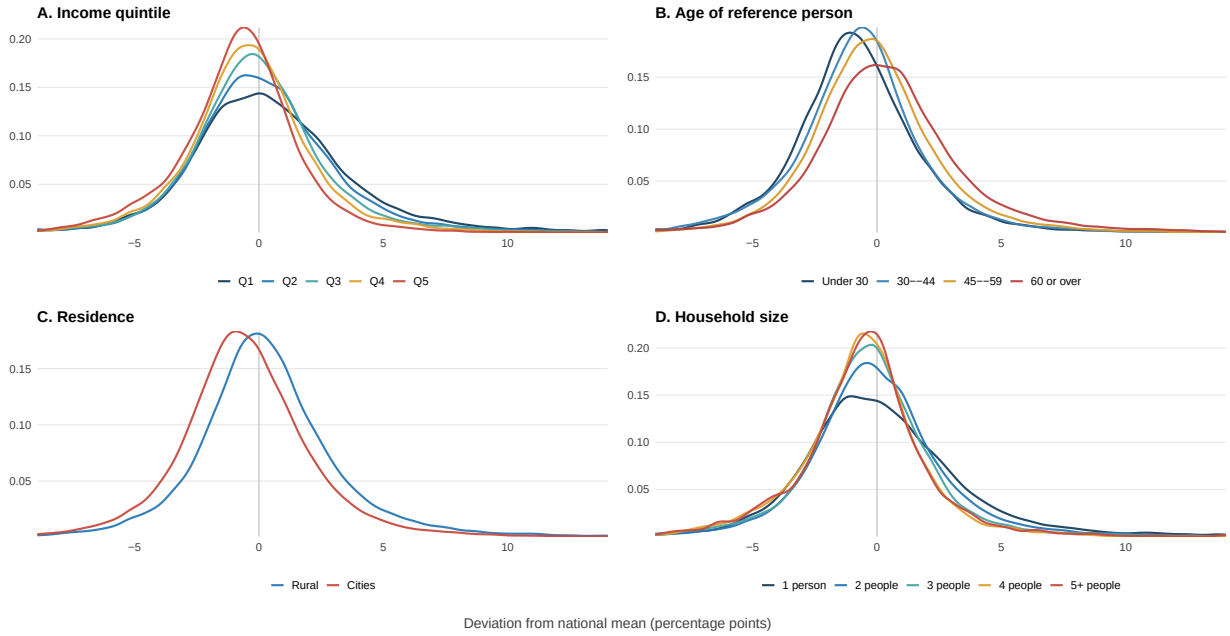
The 9.1% share is close to the 11.6% reported by O'Flaherty (2026a) for US grocery inflation. The two estimates are conceptually aligned because both compare total variance with the variance remaining within fully crossed demographic cells. In O'Flaherty's setting, the within-cell compo-

nent also captures barcode-level differences in products purchased and substitution. In our setting, all households in a country face common COICOP price changes, so the within-cell component reflects differences in expenditure shares across product categories.

2.2 Group-Level Inflation Differentials

Observable characteristics explain only a limited share of household-level dispersion, but they can still shift the location and tails of the inflation distribution. Figure 6 compares the weighted distributions of household inflation across four classifications in 2022. The curves overlap substantially, but their locations and tails differ in economically meaningful ways. The 90th percentile of inflation relative to the national mean is 4.7 percentage points for Q1 and 2.0 points for Q5. The corresponding 90th-percentile deviations are 4.2 points for households whose reference person is aged 60 or over, against 2.3 points for those under 30; 3.9 points in rural areas, against 2.7 points in cities; and 4.2 points for one-person households, against 2.1 points for four-person households. Observable groups therefore shift both the location and the tails of household inflation even though they do not separate households into distinct distributions.

Figure 6: Household inflation distributions by demographic group, 2022



Notes: Weighted kernel densities of annual household inflation relative to the survey-weighted national mean. The common sample contains 152,440 households in 18 euro-area countries. Italy is excluded because household income is unavailable for the common complete-case analysis; Portugal is excluded because its expenditure data cannot separate rents. Household inflation uses fixed 2020 HBS expenditure baskets and common national price changes at the three-digit ECOICOP1 level, excluding actual and imputed rents. National distributions are pooled using annual HICP country expenditure weights. Income quintiles are defined nationally; age refers to the household reference person. The rural group combines intermediate-density and rural areas so that every country contributes to both residence groups. All panels use the same sample, horizontal range, and kernel bandwidth. The thin vertical line marks the national mean. Sources: Eurostat HICP, HBS 2020 microdata, authors' calculations.

Group averages provide a compact measure of these systematic exposure differences, but they are much narrower than the underlying household distribution. In October 2022, the 2.2-point gap between the mean inflation rates of the first and fifth income quintiles was about one-third of the 6.5-point individual P20–P80 range reported above.

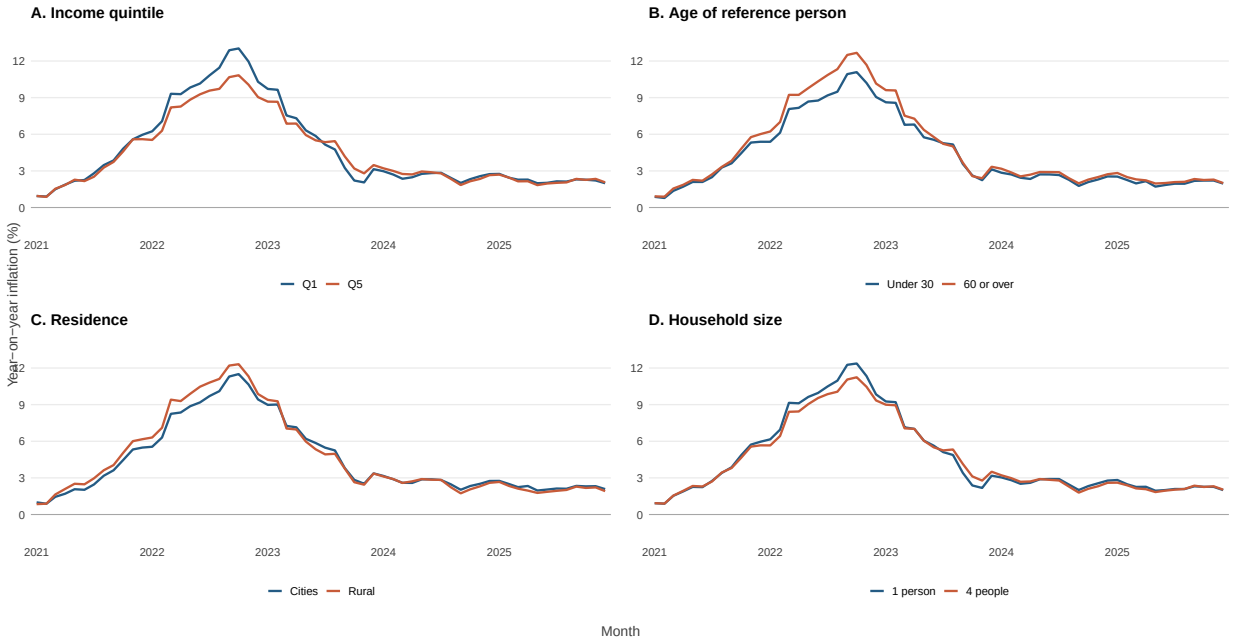
The inflation gap between households in the bottom and top 20 percent of their national income distributions reached 1.2 percentage points in 2022, its highest annual value in our series since 2000.[5] Figure 7 traces the monthly dynamics behind this annual result. From January 2021 through September 2021, the gap remained at or below 0.25 percentage point in eight of nine months, consistent with the low-inflation environment. It then rose from 0.1 percentage point in September 2021 to 2.2 points in October 2022 before narrowing as energy inflation receded. This time profile is consistent with recent evidence from Bobasu et al. (2023), although differences in weighting and country coverage may affect the measured size of the gap.

Income is not the only systematic dimension of inflation exposure. Figure 7 compares monthly inflation across four household characteristics on a common sample. The gaps peak during the energy shock: inflation is 2.2 points higher for Q1 than Q5 households, 1.8 points higher for households whose reference person is aged 60 or older than for those under 30, and 1.3 points higher in rural areas than in cities. The largest gap between one- and four-person households is 1.2 points. All four differences subsequently narrow, showing that these gradients are concentrated in the energy-shock period rather than persistent throughout the sample.

The residence-area gradient follows a similar time profile and is consistent with differences in the structure of energy demand. Rural households tend to spend a larger share of their budget on transport fuels and home energy, which raises their exposure when energy prices accelerate and makes their measured inflation fall faster when those price pressures unwind. Gautier and Montornes (2024) obtain similar results for the four largest euro-area countries under slightly different assumptions.

[5] Appendix H presents the corresponding long-run price indices.

Figure 7: Inflation by household characteristics in the euro area (year on year, in %)



Notes: Year-on-year household inflation from January 2021 to December 2025. Each panel compares two groups within the same complete-case sample of 152,440 households in 18 euro-area countries. Italy is excluded because household income is unavailable for the common complete-case analysis; Portugal is excluded because its expenditure data cannot separate rents. Household inflation uses fixed 2020 HBS expenditure baskets and common national price changes at the three-digit ECOICOP1 level, excluding actual and imputed rents. National group means use HBS survey weights and are aggregated using annual HICP country weights normalized over the 18 countries. Income quintiles are defined nationally; age refers to the household reference person. The rural group combines intermediate-density and rural areas, ensuring that Malta and every other retained country contributes to both residence series.

Sources: Eurostat HICP and HBS, authors' calculations.

2.3 Product Contributions During the 2022–2023 Energy Shock

The euro-area aggregate masks large, offsetting product contributions. Table 1 first decomposes the individual P20–P80 range. In 2022, this range reaches 4.8 percentage points. Vehicle-operation expenditure contributes 2.8 points, making differences in exposure to vehicle-operation costs, particularly motor fuels, the main source of individual dispersion; household energy contributes another 2.3 points. The concentration of the P20–P80 range in these energy-related products shows that households with similar observed characteristics can still face very different inflation because their heating and mobility needs differ.

The Q1–Q5 decomposition turns to the group-based indicator commonly used by statistical institutes to summarize inflation heterogeneity across the income distribution. It answers a narrower question: which products shift average inflation between the bottom and top income quintiles? This gap can be interpreted as a weighted covariance between relative differences in expenditure shares and product-level inflation. In 2022, household energy contributes 1.9 points and food 0.7 point to the 1.2-point aggregate gap. Motor fuels work in the opposite direction: vehicle-operation expenditure contributes -0.4 point because higher-income households devote a larger share of expenditure to private transport. Other transport and other products contribute another -0.3 and -0.5 point. Thus, vehicle-operation costs, especially motor fuels, substantially widen dispersion across individual households while partly narrowing the average Q1–Q5 gap. In 2023, the positive food contribution is again offset by other product groups, leaving a zero aggregate gap. Yet the individual P20–P80 range remains 2.6 points in 2023. This divergence shows directly that a zero gap between income-group means can coexist with substantial inflation dispersion across households.

Table 1: Product-group contributions to individual P20–P80 inflation dispersion and Q1–Q5 inflation inequality in the euro area, 2021–2023

		Inflation inequality					
		Individual P20–P80			Income Q1–Q5		
Product group	Weight (2022, %)	2021	2022	2023	2021	2022	2023
Food and non-alcoholic beverages	17.8	-0.2	-0.1	1.3	0.1	0.7	0.7
Alcoholic beverages, tobacco and narcotics	4.6	0.1	-0.4	-0.1	0.1	0.1	0.2
Clothing and footwear	5.6	-0.1	-0.1	0.1	0.0	0.0	-0.1
Housing maintenance and furnishings	8.0	0.1	0.1	0.4	-0.1	-0.3	-0.2
Water, electricity, gas and other fuels	6.8	0.1	2.3	0.7	0.4	1.9	0.1
Operation of personal transport equipment	6.3	1.1	2.8	0.0	-0.3	-0.4	0.0
Transport (excluding vehicle operation)	9.6	0.2	-0.4	-0.1	-0.1	-0.3	-0.2
Other	41.1	0.3	0.6	0.4	-0.1	-0.5	-0.6
Total	**100.0**	**1.7**	**4.8**	**2.6**	**0.1**	**1.2**	**0.0**

Notes. The first block decomposes the P80 minus P20 range of annual household inflation, in percentage points, using the product contributions of the households that define each national endpoint. National contributions are averaged with annual HICP country weights over the 19-country household sample; Portugal is excluded. Italy enters without an income filter. The second block decomposes the Q1 minus Q5 gap for all 20 euro-area countries. Annual rates compare annual-average indices with the preceding year. Product weights are 2022 EA20 expenditure shares. Rows are rounded separately, so their displayed sum may differ from the displayed total. Sources: Eurostat, HBS 2020 microdata, authors' calculations.

2.4 Cross-Country Heterogeneity During the 2022–2023 Energy Shock

Cross-country differences reflect variation in energy mixes, household consumption patterns, and fiscal responses. These factors shaped how the energy-price shock, which began in 2021 and intensified, especially in natural gas markets, after Russia's invasion of Ukraine, translated into household inflation.

Table 2 documents this cross-country heterogeneity by reporting the individual P20–P80 inflation range and the annual Quintile 1–Quintile 5 inflation gap. Individual dispersion peaks in 2022 in most countries and is especially large in Estonia (10.7 points), Lithuania (10.1), Belgium (8.0), Italy (7.7), and Latvia (7.6), compared with 4.8 points for the 19-country aggregate. The Q1–Q5

gap is largest in 2022 in most countries: it exceeds 3 points in Estonia, Latvia, Lithuania, and the Netherlands and reaches 2.9 points in Ireland. Several countries subsequently record a negative gap in 2023 as energy-price pressures recede, while the euro-area gap falls from 1.2 points in 2022 to zero in 2023. This cross-country pattern in the Q1–Q5 inflation gap is qualitatively similar to Claeys et al. (2022), who also find that low-to-high-income inflation gaps are larger in countries with higher aggregate inflation.

The same interaction between price dynamics and expenditure shares operates across countries. A price increase in a product category generates inflation inequality only to the extent that the affected category has different expenditure weights across household groups. Large aggregate shocks may generate limited inequality when price increases fall on categories with similar weights across quintiles.

Table 2: Individual P20–P80 inflation ranges and Q1–Q5 inflation inequality in the euro area, 2021–2023

		Inflation inequality					
		Individual P20–P80			Income Q1–Q5		
Country	Weight (2022, %)	2021	2022	2023	2021	2022	2023
Germany	28.1	1.4	4.3	1.8	-0.4	1.4	0.5
France	20.4	1.5	2.6	2.2	0.4	0.2	0.2
Italy	16.3	1.6	7.7	2.1	0.1	1.6	0.5
Spain	11.0	2.5	4.0	3.6	0.8	0.9	-0.7
Netherlands	5.4	2.0	6.8	3.7	0.0	3.1	-1.2
Belgium	4.0	3.3	8.0	7.3	1.3	2.3	-2.7
Austria	3.3	1.2	3.8	2.2	-0.5	0.0	0.5
Portugal	2.2	–	–	–	0.8	0.2	-0.9
Greece	2.2	2.1	4.7	2.7	0.3	1.2	-0.6
Finland	1.9	1.9	3.7	2.1	-0.2	-0.9	0.4
Ireland	1.5	2.0	5.0	3.1	0.5	2.9	1.1
Slovakia	0.8	1.9	2.6	2.6	-0.9	0.1	0.5
Croatia	0.7	1.2	2.7	2.5	0.0	0.2	-0.3
Lithuania	0.6	2.2	10.1	4.4	0.2	3.3	0.4
Slovenia	0.4	1.4	2.4	2.3	0.0	0.4	1.5
Luxembourg	0.3	2.6	4.9	2.8	0.1	0.2	-0.1
Latvia	0.3	1.5	7.6	3.8	-0.2	4.5	2.5
Estonia	0.2	3.0	10.7	3.4	0.6	4.8	1.3
Cyprus	0.2	1.5	3.0	1.7	0.0	0.8	0.2
Malta	0.1	1.9	2.2	2.2	-0.2	0.3	0.6
Euro area	**100.0**	**1.7**	**4.8**	**2.6**	**0.1**	**1.2**	**0.0**

Notes. The first block is the P80 minus P20 range of annual household inflation, in percentage points, using national HBS survey weights. Portugal is unavailable at the required household-product detail and is shown as dashes; the euro-area row averages national endpoints over the other 19 countries using annual HICP country weights. The second block is the Q1 minus Q5 gap for all 20 countries. Annual rates compare annual-average indices with the preceding year. The displayed country weights are 2022 EA20 HICP shares. Sources: Eurostat, HBS 2020 microdata, authors' calculations.

These decompositions describe realized inflation gaps, which jointly reflect household baskets, national price dynamics, and policy interventions; they do not identify the separate contribution of fiscal policy. The next section holds household baskets fixed and compares observed prices with

counterfactual price paths that remove household price policies.

3 Contributions of Household Price Policies to Inflation Inequality

This section isolates the direct contribution of household price policies to average inflation and inflation inequality. We first describe the interventions and their fiscal scale, then construct monthly price paths without the policies and estimate their euro-area and country-level effects.

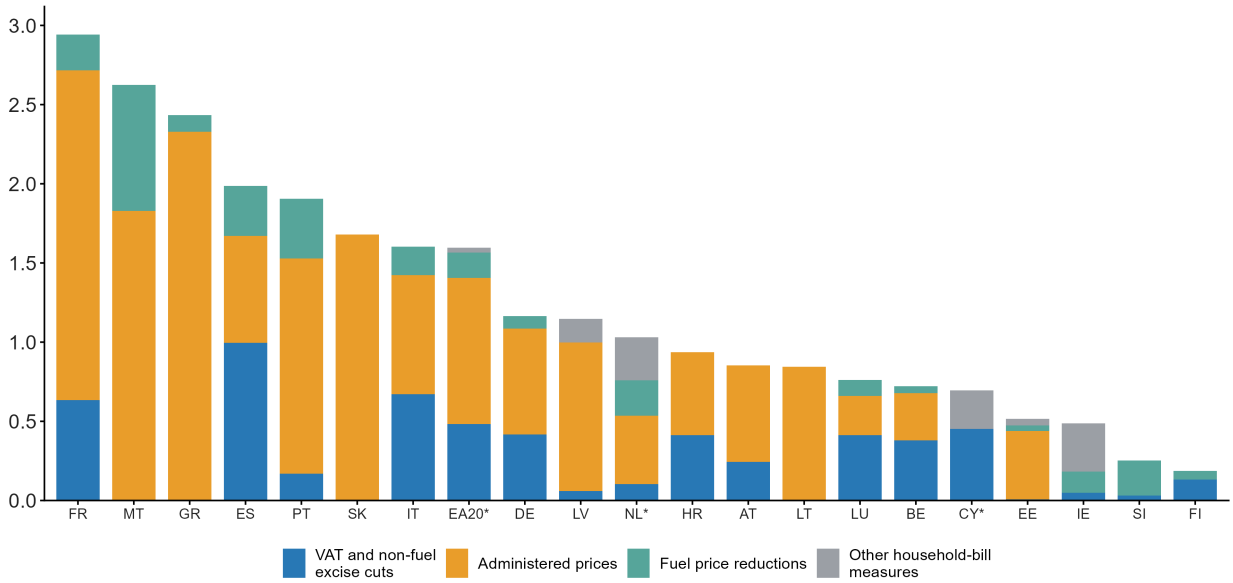
3.1 Household Price Policies during the Energy Crisis

During the 2021–2023 energy crisis, euro-area governments used electricity and gas price caps, VAT and energy-tax reductions, fuel rebates, and support applied directly to household energy bills (Arregui et al., 2022, Dao et al., 2023, García-Miralles, 2023). Because these measures alter retail prices, they enter household price indices directly. The inventory primarily covers energy measures, together with temporary food-price policies.

We combine the Bruegel energy-policy dataset (Sgaravatti et al., 2021) with national sources and organize the measures into four instrument groups: VAT and non-fuel excise reductions; administered prices; subsidies, rebates, and other fuel-price reductions; and other household-bill measures. Appendix Table I.1 reports the country-level inventory.

Figure 8 summarizes the scale and composition of the response. The documented gross fiscal cost is EUR 220.1 billion, at least 1.6 percent of euro-area GDP, of which EUR 118.1 billion (53.7%) is associated with untargeted measures. This predominance of broad-based price support is consistent with the Bruegel and OECD inventories (Hemmerlé et al., 2023, Sgaravatti et al., 2021). Administered-price policies account for the largest instrument share. Fiscal cost measures the scale of the intervention, not its household incidence or the size of the price counterfactual.

Figure 8: Fiscal Costs of Household Price Policies (Percent of 2022 GDP)



Notes: Bars sum documented costs of the household price policies active in the counterfactual during 2021–2023, counting each shared budget envelope once. Fuel price reductions comprise all road-fuel (CP0722) relief, including excise-duty reductions, regulated-price support, and direct subsidies; the VAT and non-fuel excise category therefore excludes road-fuel policies. Other household-bill measures comprise remaining instruments and shared envelopes with no identified instrument split. An asterisk indicates an unidentified instrument split. Budget reference periods vary across schemes and do not uniformly cover 2021–2023; costs are not restricted to amounts received by households. EA20 divides summed costs by summed nominal 2022 GDP.
Sources: Bruegel, national official sources, Eurostat; authors' calculations using the fiscal-cost registry underlying the policy tables.

3.2 Constructing Counterfactual Price Indices

We compare observed, policy-inclusive retail prices with counterfactual price paths that remove the covered household price policies. This comparison holds consumption quantities fixed and allows only prices to differ between the "with-policy" and "without-policy" scenarios. It is therefore an accounting counterfactual rather than a behavioral one: it does not capture quantity adjustments, such as the observed decline in residential energy consumption during the crisis. For each instrument, we use the available national evidence, including statutory rules, published reference paths, administrative data, or comparable-country ratios.

Counterfactual component price paths are first constructed at the four-digit COICOP level by

replacing each observed HICP component index with its counterfactual counterpart; they are then aggregated to the common HBS support described in Section 1. Let $p_{j,t}$ denote the observed HICP component index for product j in month t, and let $\widetilde{p}_{j,t}$ denote the corresponding counterfactual index. Here t indexes calendar months, corresponding to the year–month pair (y, m) used above. Let $r_{j,t}$ denote the observed retail price per unit and $\widetilde{r}_{j,t}$ its counterfactual counterpart, expressed in the same monetary units. For each simulated policy, we apply the implied price ratio to the observed component index[6]:

$$\widetilde{p}_{j,t} = p_{j,t} \frac{\widetilde{r}_{j,t}}{r_{j,t}}. \tag{6}$$

VAT and Non-Fuel Excise-Tax Reductions. For VAT reductions, the observed retail price includes the temporary tax rate. We recover the counterfactual price by removing that rate and applying the standard rate:

$$\widetilde{r}_{j,t} = r_{j,t} \frac{1 + \tau^0_{j,t}}{1 + \tau^1_{j,t}}, \tag{7}$$

where $\tau^0_{j,t}$ is the standard VAT rate and $\tau^1_{j,t}$ is the temporary rate. Non-fuel ad valorem excises follow the same ratio logic. The statutory rate changes are available for all VAT measures in the inventory—Belgium, Croatia, Cyprus, Finland, Germany, Ireland, Italy, Luxembourg, the Netherlands, Portugal, and Spain; when a measure covers only part of an HICP component or a subset of households, we adjust the statutory ratio for the documented coverage rather than imputing a missing tax rate. For Dutch energy-tax relief, published with-policy bills B^1_k and statutory support amounts S_k imply $\widehat{\rho}_k = 1 + S_k / B^1_k$.

Administered Prices. For regulated-price shields and caps, we use an external reference path where available, such as a commodity-price benchmark or a tariff published by the national regulator. If $T^{\text{obs}}_{j,t}$ denotes the regulated tariff actually applied and $T^{\text{ref}}_{j,t}$ the tariff that would have applied without the intervention, the counterfactual retail price is:

$$\widetilde{r}_{j,t} = r^{\text{ref}}_{j,t} = r_{j,t} \frac{T^{\text{ref}}_{j,t}}{T^{\text{obs}}_{j,t}}. \tag{8}$$

Country-specific reference paths or policy wedges are available for Austria, Germany, the Netherlands, Italy, Spain, France, Portugal, and Slovenia (Appendix Table I.2). France uses published no-shield tariff paths; Austria applies its statutory electricity-brake formula to monthly E-Control prices; and Italy derives a period-specific aggregate wedge from ARERA social-bonus funding

[6]When several policies apply to the same product in the same country and month, the R scripts combine their counterfactual-to-observed price ratios by adding their deviations from one to a baseline ratio of one.

and domestic energy expenditure. The remaining country-specific ratios use official program rules or administrative settlement data. For Dutch electricity and gas price caps, published bill and support information determines the counterfactual ratio. When both bills are reported, as for district heating, we use $\widehat{\rho}_k = B_k^0 / B_k^1$, where B_k^0 is the no-policy bill.

Subsidies, Rebates, and Other Fuel-Price Reductions. For unit subsidies and fuel rebates, we add the statutory absolute price wedge to the observed monthly price path or, without a retail unit-price series, calibrate it in HICP component-index units. Evidence from Drolsbach et al. (2023) suggests that temporary fuel-tax reductions and discounts were largely passed through to consumer prices in 2022 in several euro-area countries, although pass-through varied over time and by fuel type. This supports treating these policies as retail-price wedges in the counterfactual construction. Under full pass-through, adding the statutory support amount $s_{j,t}$ to the observed unit price gives:

$$\widetilde{r}_{j,t} = r_{j,t} + s_{j,t}. \tag{9}$$

For transport-fuel rebates, $s_{j,t}$ is recovered from the statutory pump discount, expressed in euros per liter, and applied to a monthly retail-fuel price $r_{j,t}^{\mathrm{M}}$, where the upright superscript M denotes the observed market price per liter corresponding to HICP component j. Following Bonnet et al. (2025), we refer to these interventions as pump-price rebates or subsidies because they reduced the tax-inclusive price paid by consumers, although their technical implementation sometimes operated through fuel excise taxation rather than through a direct transfer to households. The no-rebate fuel index is obtained by rescaling the observed HICP transport-fuel component:

$$\widetilde{p}_{j,t} = p_{j,t} \frac{r_{j,t}^{\mathrm{M}} + s_{j,t}}{r_{j,t}^{\mathrm{M}}}.$$

These transport-fuel policies are applied only to fuels for personal transport equipment. The product decompositions nevertheless report the broader vehicle-operation aggregate; its remaining components retain their observed price paths unless they are affected by a separately identified policy.

The inventory identifies statutory unit rebates or excise changes for motor fuels in Belgium, Cyprus, Estonia, France, Germany, Greece, Ireland, Italy, Luxembourg, the Netherlands, Portugal, and Spain, and for heating fuels in Cyprus and Slovenia. We use these source values whenever the retail-price denominator and policy coverage allow a direct conversion into a counterfactual ratio. The reconstructed ratio, $\widehat{\rho}_k$, is the unit price without the policy divided by the price paid under the policy. If the source provides an effective unit support s_k and a representative with-policy unit price

r_k^1, then $\widehat{\rho}_k = 1 + s_k/r_k^1$; if it instead provides the no-policy unit price r_k^0, then $\widehat{\rho}_k = r_k^0/(r_k^0 - s_k)$. We adjust s_k for documented coverage or pass-through.

When the source does not provide a usable unit amount or rate, price denominator, or coverage weights, we use the comparable-policy imputation described below.

Other Household-Bill Measures. This residual category contains household-bill interventions that do not fit the three preceding economic categories: VAT and non-fuel excise reductions, administered prices, or fuel-price reductions. We construct their counterfactual ratios from the source-based bill or support information available for each measure.

Comparable-Policy Imputation. When a direct reconstruction lacks a required input, irrespective of the instrument category, we estimate a counterfactual-to-observed price ratio from comparable euro-area policies. For measure k in recipient country c, $\bar{\rho}^{\mathrm{D}}_{k,-c}$ denotes the mean source-based ratio among policies matched by product group and retail-price mechanism, excluding country c. We set $\widehat{\rho}_k = \bar{\rho}^{\mathrm{D}}_{k,-c}$, first averaging multiple measures within each donor country and then weighting donor countries equally. We record these imputations separately from direct reconstructions. Appendix Table I.3 compares the full specification with direct reconstructions only and with alternative conditional-benchmark wedges. For the incomplete-input measures in Cyprus, Ireland, and Latvia, the Netherlands is the sole donor.

We then replace observed component indices with these counterfactual paths and apply the same RAS-calibrated weights and household-group, national, and euro-area aggregation procedures as in Section 1.

3.3 Effects by Instrument and Targeting Status

Table 3 provides the first application of the counterfactual indices by decomposing annual effects across the four economic instrument categories and by targeting status. Effects are observed minus counterfactual outcomes, so negative entries indicate that the policies reduce inflation dispersion or the Q1–Q5 gap.

For individual dispersion, administered-price policies reduce the P20–P80 range by 1.1 points in 2022 and 0.3 point in 2023. Untargeted policies reduce it by 1.2 and 0.5 points, while targeted policies reduce it by 0.6 and 0.1 points. The corresponding Q1–Q5 results are smaller: administered-price policies reduce the gap by 0.4 point in both 2022 and 2023. Targeted policies reduce it by 0.3 point in 2022 and 0.1 point in 2023; untargeted policies reduce it by 0.4 and 0.5 points, respectively.

The category effects are not additive because each counterfactual recomputes the households at the national percentile endpoints. In 2023, the displayed instrument effects on P20–P80 sum to -0.5 point, while removing all policies jointly gives -0.7 point. In 2021, the category effects round individually to zero, while their joint effect rounds to -0.1 point.

The 2022 magnitude is economically large. The observed annual P20–P80 range of 4.8 points and the -1.8-point policy effect imply a no-policy range of 6.6 points. Household price policies therefore compress annual individual inflation dispersion by about 27% relative to that counterfactual. Appendix Table I.3 reports an effect of -1.1 points using direct reconstructions only and effects between -1.7 and -1.8 points across the alternative conditional-benchmark specifications.

The fuel-price-reduction category includes all road-fuel policies, including excise-duty reductions, regulated-price support, and direct subsidies. The VAT and non-fuel excise category therefore excludes road-fuel policies. Italian social energy bonuses and electricity-system charge relief are classified as administered-price policies because they directly reduce regulated bill components. Other household-bill measures contain only interventions that do not belong to these first three categories; shared envelopes enter this residual category when no unique instrument split is available. Of the EUR 220.1 billion total, EUR 15.0 billion has no identified targeting status and therefore enters Total but neither targeting row.

Table 3: Annual effects of household price-policy categories on individual P20–P80 inflation dispersion and Q1–Q5 inflation inequality, 2021–2023

		Effect of price policies on					
		Individual P20–P80			Income Q1–Q5		
Policy category	Total fiscal cost (EUR bn)	2021	2022	2023	2021	2022	2023
Instrument classification							
VAT and non-fuel excise cuts	66.7	0.0	-0.5	0.0	0.0	-0.2	-0.1
Administered prices	127.0	0.0	-1.1	-0.3	0.0	-0.4	-0.4
Fuel price reductions	21.9	0.0	-0.2	-0.2	0.0	0.1	0.0
Other household-bill measures	4.4	0.0	0.0	0.0	0.0	0.0	0.0
Total	**220.1**	**-0.1**	**-1.8**	**-0.7**	**-0.1**	**-0.6**	**-0.6**
Targeted vs Untargeted							
Untargeted price measures	118.1	-0.1	-1.2	-0.5	0.0	-0.4	-0.5
Targeted price measures	87.0	0.0	-0.6	-0.1	0.0	-0.3	-0.1
Total	**220.1**	**-0.1**	**-1.8**	**-0.7**	**-0.1**	**-0.6**	**-0.6**

Notes. Observed minus counterfactual effect on the P80 minus P20 range of annual household inflation or on the Q1–Q5 gap, in percentage points; negative values indicate a reduction. National percentile endpoints use HBS survey weights and are averaged with annual HICP country weights over the 19-country household sample; Portugal is excluded from the first block, and Italy enters without an income filter. Fuel price reductions include all road-fuel relief, including excise-duty reductions, regulated-price support, and direct subsidies; the VAT and non-fuel excise category excludes road-fuel measures. Category effects need not sum to Total because policies interact. Targeting classifies measures, not individual eligibility. Fiscal costs are total documented policy envelopes across the 20 countries, in EUR billions; budget periods may span several years. EUR 15.0 billion with unidentified targeting status enters Total but neither targeting row. Shared envelopes without a unique instrument split are assigned to Other household-bill measures. Fiscal-cost rows are rounded independently and may not sum exactly to the displayed total. Fiscal costs refer to entire schemes affecting household prices, not amounts received exclusively by households. Sources: Eurostat, HBS 2020 microdata, national sources, authors' calculations.

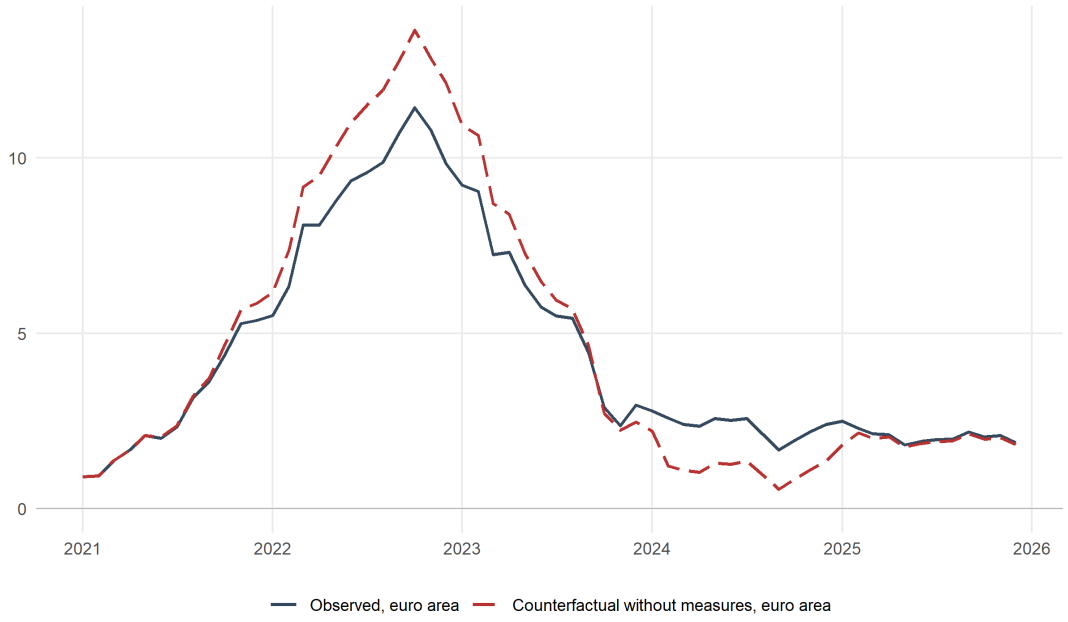
3.4 Euro-Area Effects on Inflation and Inflation Inequality

Household price policies changed both the level and the timing of inflation during the energy crisis. We isolate their direct effect on consumer prices. Observed minus counterfactual euro-area annual inflation is -0.1 percentage point in 2021, -1.7 points in 2022, and -0.6 point in 2023 (Figure 9). The 2022 estimate is consistent in magnitude with Dao et al. (2023), who find that such fiscal policies reduced euro-area inflation by 1–2 percentage points in 2022 under different modeling assumptions. Counterfactual inflation lies above observed inflation during the main phase of the energy shock, with the largest difference around the inflation peak in late 2022. VAT and

non-fuel excise reductions, fuel-price reductions, price shields, and regulated-price caps therefore compressed the peak.

A reduction in the price level need not lower inflation in every subsequent year. When a temporary policy expires or its price discount shrinks, observed prices rise relative to the lower comparison base created by the policy. Observed annual inflation can then exceed counterfactual inflation even if the observed price level remains no higher than its no-policy counterpart. Positive annual policy effects therefore can reflect the withdrawal of earlier support and base effects, rather than a higher price level caused by the policy.

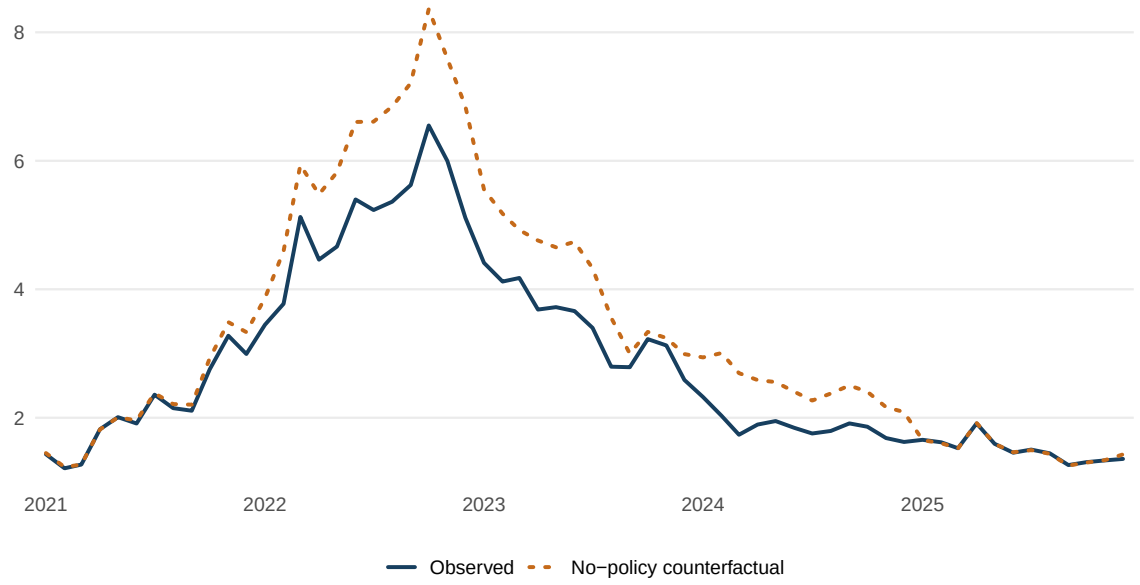
Figure 9: Household price policies: observed and counterfactual euro-area inflation (year on year, in %)



Notes: The counterfactual series removes household price policies. The figure reports year-on-year inflation for all 20 euro-area countries, aggregated with annual HICP country weights, from January 2021 to December 2025. Policy effects are defined as observed minus counterfactual inflation. The plotted indices use the same product-to-country-to-euro-area aggregation as the annual estimates in Table 4.
Sources: Eurostat HICP; Bruegel energy-policy dataset; authors' calculations.

Household price policies also changed the dispersion of inflation across households. Figure 10 traces the monthly P20–P80 range under the observed policy regime and in the no-policy counterfactual.

Figure 10: Observed and counterfactual P20–P80 household-inflation range, 2021–2025



Notes: Each line is the difference, in percentage points, between the P80 and P20 of monthly year-on-year household inflation from January 2021 to December 2025. The solid line is observed; the dashed line removes the household price policies implemented during 2021–2023. Their effects can persist into 2024 through the year-on-year comparison base. Percentile endpoints are first estimated within each country using HBS survey weights and then averaged using annual HICP country weights. The sample comprises 19 euro-area countries; Portugal is excluded. Italy enters without an income filter. Income transfers and firm-only policies are outside the counterfactual.
Sources: Eurostat HICP and country weights; household budget surveys; Bruegel energy-policy dataset; authors' calculations.

The monthly P20–P80 range averages 5.1 points in 2022, compared with 6.3 points without the policies, so the policies reduce this monthly measure by 1.3 points on average. This statistic averages the P20–P80 range calculated separately in each month. The 1.8-point effect in Table 3 instead calculates P20–P80 after first constructing each household's annual-average inflation rate; the two need not coincide because percentiles are nonlinear. In 2023, policies reduce both measures by 0.7 point. The monthly series shows when this compression emerged: the counterfactual range moves above the observed range as support expands in 2022 and remains generally wider through 2024 because of policy timing and inflation base effects; the two series converge in 2025. This pattern is consistent with microsimulation evidence that the European fiscal response reduced distributional losses, although broad price policies were less targeted than income support (Amores et al., 2025).

The product decomposition confirms that household energy drives the distributional effect of

these policies (Appendix Table I.4). Policies affecting vehicle operation partly offset the reduction in individual dispersion, contributing 2.0 points to the P20–P80 effect. The energy shock widened inequality, while price policies cushioned it.

3.5 Differences Across Countries

Table 4 reports the annual effects of each country's household price policies on average inflation, the individual P20–P80 inflation range, and the Quintile 1–Quintile 5 inflation gap, together with their fiscal costs. The costs describe the scale of the schemes rather than household incidence; an asterisk identifies envelopes that also include business beneficiaries.

Table 4: Household price policies: fiscal cost and effects on average inflation, individual P20–P80 inflation dispersion, and Q1–Q5 inflation inequality, 2021–2023

Country	Fiscal cost (% 2022 GDP)	Effect of price policies on								
		Average inflation			Individual P20–P80			Income Q1–Q5		
		2021	2022	2023	2021	2022	2023	2021	2022	2023
Germany	1.2*	0.0	-0.7	-1.5	0.1	-0.5	-1.1	0.0	-0.3	-1.2
France	2.9	-0.1	-2.7	-0.6	-0.1	-2.7	-1.1	0.0	-1.1	-0.5
Italy	1.6	-0.2	-2.2	0.8	-0.3	-2.8	-0.4	-0.1	-0.5	0.0
Spain	2.0*	-0.5	-2.3	-0.5	-0.5	-2.2	-0.2	-0.3	-0.9	-0.4
Netherlands	1.0*	0.0	-2.5	-0.5	0.0	-2.5	0.3	0.0	-1.2	-0.3
Belgium	0.7*	0.0	-2.4	0.2	0.0	-3.1	-0.2	0.0	-1.1	0.0
Austria	0.9*	0.0	-0.2	-0.8	0.0	-0.2	-0.7	0.0	-0.1	-0.4
Portugal	1.9	0.0	-1.0	-2.5	–	–	–	0.0	-0.5	-1.7
Greece	2.4*	-0.3	-0.9	0.3	-0.3	-0.6	-0.3	-0.2	-0.3	0.0
Finland	0.2	0.0	-0.4	-0.4	0.0	-0.5	0.0	0.0	0.2	0.1
Ireland	0.5*	0.0	-0.9	0.0	0.0	-1.1	-0.3	0.0	-0.5	-0.1
Slovakia	1.7	0.0	0.0	-3.6	0.0	0.0	-2.7	0.0	0.0	-2.0
Croatia	0.9	0.0	-1.2	0.4	0.0	-0.7	-0.4	0.0	-1.1	0.5
Lithuania	0.8	0.0	-0.4	-0.1	0.0	-0.5	0.0	0.0	-0.4	-0.1
Slovenia	0.3	0.0	-1.0	-0.1	0.0	-0.6	-0.8	0.0	-0.1	-0.6
Luxembourg	0.8*	0.0	-0.7	-0.6	0.0	-0.6	-0.4	0.0	-0.3	-0.2
Latvia	1.1	0.0	-1.5	-0.7	0.0	-1.8	-0.3	0.0	-1.7	-0.5
Estonia	0.5*	-0.2	-1.1	0.4	-0.2	-1.8	-0.1	-0.1	-0.9	0.4
Cyprus	0.7	-0.1	-0.6	0.6	0.0	-0.4	-0.1	0.0	-0.2	0.2
Malta	2.6*	0.0	-0.9	0.0	0.0	-0.1	0.1	0.0	-0.1	0.0
Euro area	**1.6**	**-0.1**	**-1.7**	**-0.6**	**-0.1**	**-1.8**	**-0.7**	**-0.1**	**-0.6**	**-0.6**

Notes. Effects are observed minus counterfactual annual inflation, in percentage points. The first block reports average inflation. The second reports the P80 minus P20 range of household inflation; national endpoints use HBS survey weights. Portugal is unavailable at the required household-product detail and is shown as dashes; the euro-area row averages national endpoints over the other 19 countries using annual HICP country weights. The third block is the Q1 minus Q5 gap for all 20 countries. Costs are scheme budgets as a percentage of 2022 GDP, counting shared envelopes once. An asterisk marks costs including business beneficiaries; the household share is not identified. A dash denotes incomplete coverage, not zero. France's fuel rebate excludes EUR 1.6 billion paid to road freight; its electricity shield also covers microbusinesses. Source periods vary (France and Lithuania: 2022); Slovak amounts are gross 2023 expenditure. Dutch quintile effects measure basket exposure, not cap eligibility. Sources: Eurostat, Bruegel, national sources, authors' calculations.

The distributional effects are heterogeneous across countries. For the individual P20–P80 range, the largest reductions in 2022 occur in Belgium (3.1 points), Italy (2.8), France (2.7), the Netherlands (2.5), and Spain (2.2); in 2023, Slovakia records the largest reduction (2.7 points). For the Q1–Q5 gap, the largest reductions in 2022 occur in Latvia (1.7 points), the Netherlands (1.2), Belgium, France, and Croatia (1.1 each), and Estonia and Spain (0.9 each). In 2023, Slovakia records the largest reduction (2.0 points), followed by Portugal (1.7), Germany (1.2), Slovenia (0.6), and France and Latvia (0.5 each). Policies increase the gap by 0.5 point in Croatia, 0.4 in Estonia, and 0.2 in Cyprus in 2023.

The Netherlands illustrates why effects on average inflation and dispersion need not have the same sign. In 2023, its policies reduce average inflation by 0.5 point but increase the individual P20–P80 range by 0.3 point because the induced price changes are uneven across household baskets.

These estimates measure basket exposure to national component-price paths, not household eligibility or behavioral responses. Fiscal costs also refer to different budget periods and may combine forecast and realized amounts. Inflation-gap effects are percentage-point changes, not monetary household gains. Taken together, the country results show that policy effects on inflation inequality arise from the interaction between policy design and household expenditure shares. Policies concentrated on products whose expenditure shares differ most across households have larger distributional effects because they change inflation more for highly exposed households. When expenditure shares are similar, the policy shifts household inflation rates by similar amounts: average inflation falls, but the P20–P80 range and Q1–Q5 gap change little. The size and timing of the price wedge also shape the annual effect. Fiscal cost records the budgetary scale of a measure, not how strongly it changes the price of a particular product or which households spend most on that product. It therefore does not by itself predict the size or direction of the inflation-gap effect.

4 Inflation Burden

Our indicators measure differences in inflation exposure across household groups, but they do not directly express the cost of inflation relative to household resources. We therefore construct the burden from HBS microdata rather than combining group-level inflation with an external aggregate consumption-to-income ratio. For household i in country c, let X_{ic} denote consumption expenditure on the non-rent products matched to HICP, Y_{ic} net household income, and $\bar{\pi}_{ic,y}$ the household's average annual inflation rate in year y. The raw household-level burden is

$$\mathcal{B}^{\text{raw}}_{ic,y} = \bar{\pi}_{ic,y} \frac{X_{ic}}{Y_{ic}}, \tag{10}$$

where $\bar{\pi}_{ic,y}$ is measured in percent, so $\mathcal{B}^{\text{raw}}_{ic,y}$ is measured in points of net income. The indicator allows both individual inflation exposure and the consumption-to-income ratio to vary across households. The expenditure numerator and inflation index cover the same matched non-rent products. This fixed-resource exposure indicator is not compensating variation: it holds 2020 expenditure and income fixed and allows neither substitution nor income adjustment.

Very small reported incomes can generate mechanically extreme expenditure-to-income ratios. We therefore winsorize $100X_{ic}/Y_{ic}$ at the survey-weighted 99th percentile within each country, while retaining the underlying household inflation rate. Denoting the resulting ratio by $\widetilde{\kappa}_{ic}$, the burden used in the estimates and its aggregation for national income quintile q are

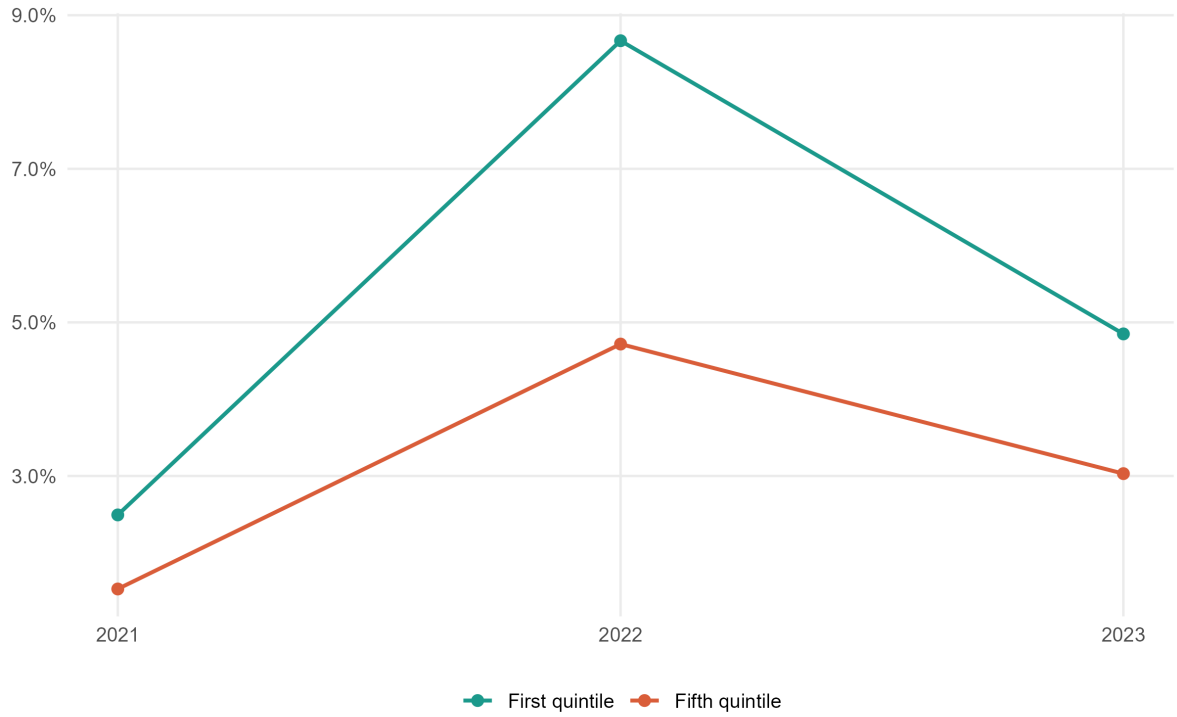
$$\mathcal{B}_{ic,y} = \bar{\pi}_{ic,y}\frac{\widetilde{\kappa}_{ic}}{100}, \qquad \mathcal{B}_{q,y} = \frac{\sum_c \sum_{i\in q(c)} a_{ic}\mathcal{B}_{ic,y}}{\sum_c \sum_{i\in q(c)} a_{ic}}, \tag{11}$$

where a_{ic} is the HBS survey weight and $q(c)$ indicates that total-net-income quintiles are defined separately within each country before households are pooled. The survey weights expand the microdata to national household populations and therefore determine the euro-area aggregation. The 2020 HBS basket, income, and expenditure observations are held fixed, while household-specific inflation is calculated for 2021–2023. For each household-year, the inflation rate is computed over the matched non-rent expenditure categories and the HBS–HICP product weights are renormalized to one. We impose no minimum matched-expenditure threshold.

Net household income enters this calculation twice: it determines the national income quintile and appears in the denominator of X_{ic}/Y_{ic}. The Q1–Q5 gradient can therefore arise partly from the mechanical expenditure-to-income gradient, even if households face the same inflation rate. We report the estimates as a descriptive fixed-resource exposure measure and do not interpret the quintile difference as the causal effect of heterogeneous inflation. Separating the inflation and expenditure-to-income components requires a common-inflation counterfactual and sensitivity checks using medians, alternative winsorization thresholds, and equivalized income.

Figure 11 uses household indices and matched non-rent expenditure. The first quintile faces a burden of 2.5 percent of net income in 2021, 8.7 percent in 2022, and 4.8 percent in 2023, compared with 1.5, 4.7, and 3.0 percent for the fifth quintile. The Q1–Q5 gap is 1.0, 4.0, and 1.8 points, respectively, calculated before rounding. The income and expenditure filters retain positive-income households; national quintiles are assigned before these burden-specific exclusions.

Figure 11: Household-level inflation burden in the euro area, by income quintile (in % of net income)



Notes: Annual household inflation times matched non-rent expenditure divided by net income; ratios are winsorized at the national survey-weighted 99th percentile. Survey-weighted means within national total-income quintiles, 18 countries excluding Italy and Portugal. Fixed 2020 resources; this is a descriptive exposure measure, not a welfare estimate. Because income determines both quintile assignment and the denominator, the Q1–Q5 gradient partly reflects differences in expenditure-to-income ratios.

Sources: HBS microdata, Eurostat HICP, authors' calculations.

5 Conclusion

This paper constructs household-specific inflation indices to measure inflation inequality in the euro area. Its central result is that dispersion across individual households is substantially wider than differences between broad household categories. At the inflation peak in October 2022, the P20–P80 range reached 6.5 percentage points, nearly three times the 2.2-point difference between the mean inflation rates of the first and fifth income quintiles. A model saturated with interactions among income, age, household size, and urbanization explains only 9.1% of household inflation variance on average; 90.9% remains within observationally similar groups. The annual product decomposition reinforces this result: vehicle-operation expenditure, especially motor fuels, accounts for 2.8 points of the 4.8-point P20–P80 range in 2022, while household energy accounts for another 2.3 points. Detailed differences in heating and mobility needs therefore generate substantial inflation heterogeneity that group averages cannot fully capture (O'Flaherty, 2026b).

Group indicators nevertheless identify systematic exposure gradients. The annual Q1–Q5 inflation gap peaked at 1.2 points in 2022, while monthly gaps reached 2.2 points for income, 1.8 points for age, and 1.3 points between rural and city households during the energy shock. Household energy widened the Q1–Q5 gap because it carries greater budget weight among lower-income households, whereas motor fuels partly narrowed that gap because private-transport expenditure is higher among richer households. The same product can thus have different implications for individual dispersion and for differences between group means.

Household price policies cushioned both dimensions of inflation inequality. In 2022, they reduced annual-average inflation by 1.7 percentage points, the individual P20–P80 range by 1.8 points, and the Q1–Q5 gap by 0.6 point. The corresponding reductions in 2023 were 0.6, 0.7, and 0.6 point. Household-energy relief accounts for most of the narrowing, while policies affecting vehicle operation partly offset the reduction in individual dispersion. These fixed-basket accounting effects depend on the products covered, the timing and size of the price wedge, and household expenditure shares.

The inflation burden provides a complementary perspective by scaling household inflation by matched non-rent expenditure relative to net income. In 2022, this burden reached 8.7% of net income for the first income quintile and 4.7% for the fifth, a gap of 4.0 points. This gradient combines differences in inflation exposure with the higher expenditure-to-income ratios of low-income households. It is therefore a descriptive fixed-resource measure rather than a welfare effect attributable solely to heterogeneous inflation.

An extension would combine household-specific price indices with tax-benefit records and balance-sheet data. This integrated framework would extend the analysis beyond consumption exposure to income indexation, transfers, debt, and wealth, providing a fuller account of changes in household purchasing power and a stronger basis for evaluating policies intended to cushion inflation's distributional effects.

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Appendix

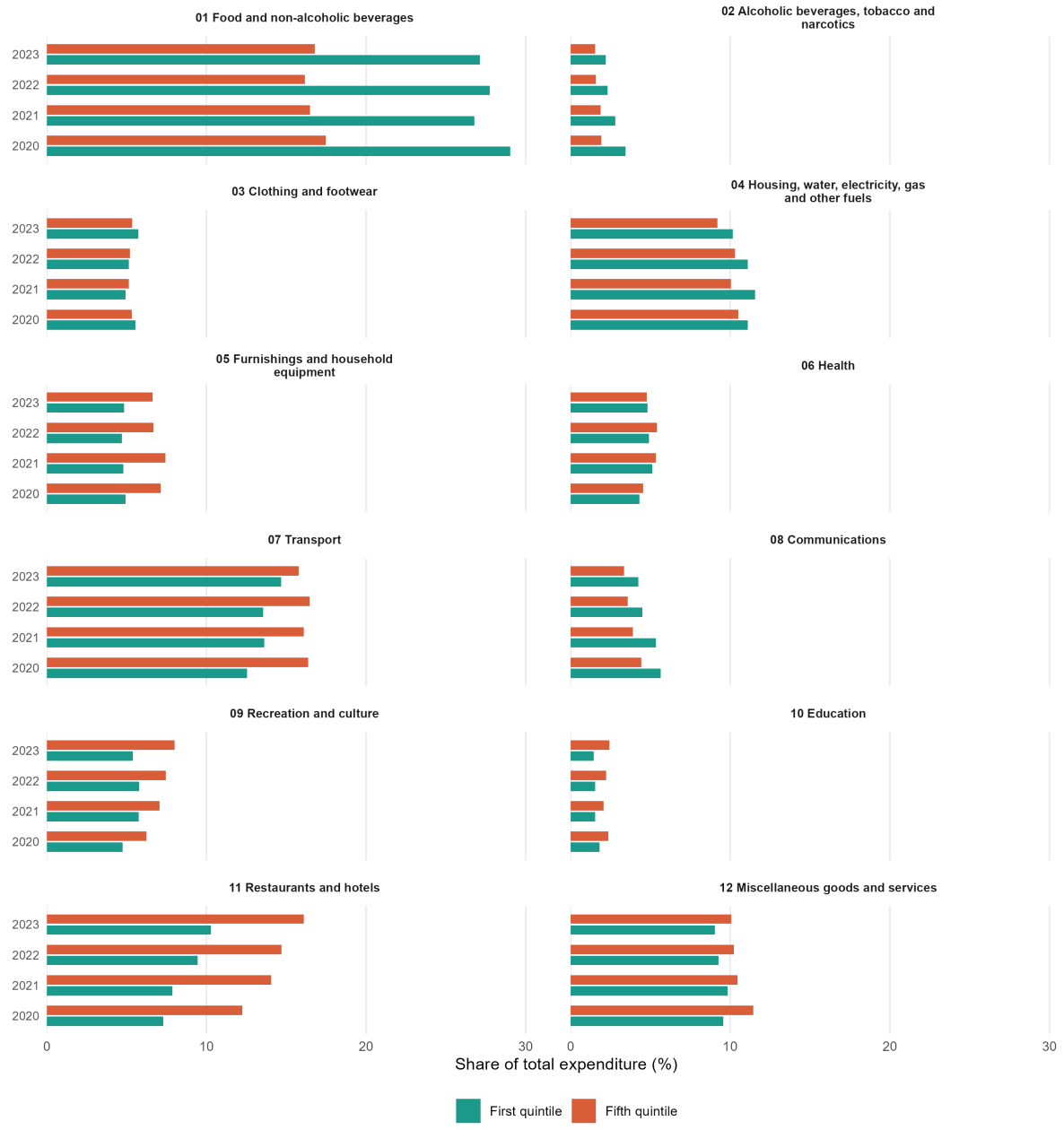
Erwan Gautier - Jérémi Montornès

(Not for publication)

October 21, 2026

A Stability of Income-Quintile Expenditure Patterns

Figure A.1: Main COICOP expenditure weights by income quintile, Spain (in % of total expenditure)



Notes: The figure reports the share of each main COICOP category in total household expenditure for the first and fifth equivalized-income quintiles. Quintiles are computed separately by year using household weights.

Sources: INE Encuesta de Presupuestos Familiares public-use microdata; authors' calculations.

B Individual Inflation Dispersion Across COICOP Digits

Table B.1 evaluates how product aggregation affects the individual P20–P80 inflation range. Household inflation is recomputed at each available COICOP digit depth from fixed 2020 HBS baskets and annual expenditure-margin RAS weights. For Malta at digit 5, relative-expenditure weights are used because household RAS does not converge. Actual and imputed rents (HBS 041 and 042) and the corresponding HICP actual-rent component are excluded before product aggregation. France, Italy and Spain use the dedicated digit-4 household files; Germany is available through digit 3, while the remaining countries use their finest available detail. Portugal is excluded because its detailed household expenditure fields are unavailable. The resulting sample therefore contains 19 countries and does not require household income.

Inflation and aggregation bias. Let $I^d_{i,y,m}$ denote the chained household index for household i at digit depth d. Inflation in 2022 is calculated from annual-average indices,

$$\pi^d_{i,2022} = 100 \left(\frac{\frac{1}{12}\sum_{m=1}^{12} I^d_{i,2022,m}}{\frac{1}{12}\sum_{m=1}^{12} I^d_{i,2021,m}} - 1 \right). \tag{12}$$

For each country, $G^d = P_{80}(\pi^d_i) - P_{20}(\pi^d_i)$, where the percentiles use HBS survey weights. Aggregation bias is $B^{d,d'} = G^d - G^{d'}$ for $d' > d$; a negative value therefore indicates that the finer product classification reveals a wider household-inflation range. Because matched coverage can vary with digit depth, the comparison can also reflect changes in product support.

Table B.1: Individual inflation dispersion across COICOP digit depths, excluding rents (2022)

	P20–P80 range (pp)				Aggregation bias (pp)			Coverage (%)			
Country	D2	D3	D4	D5	D2–D3	D2–D4	D2–D5	D2	D3	D4	D5
AT	3.4	3.8	5.1	5.3	-0.5	-1.7	-1.9	100.0	100.0	99.8	97.5
BE	6.7	8.1	8.9	8.7	-1.4	-2.1	-2.0	100.0	100.0	99.7	98.1
CY	2.6	3.0	3.2	3.4	-0.3	-0.5	-0.8	100.0	100.0	99.8	99.0
DE	3.4	4.3	–	–	-0.9	–	–	100.0	100.0	–	–
EE	9.7	10.8	10.8	11.3	-1.1	-1.1	-1.6	100.0	99.9	99.6	97.6
EL	3.2	4.7	4.9	5.1	-1.5	-1.7	-1.8	100.0	99.7	99.5	99.3
ES	2.9	4.0	4.5	–	-1.2	-1.6	–	100.0	99.9	98.1	–
FI	2.8	3.7	4.4	4.5	-0.9	-1.6	-1.7	100.0	99.7	98.2	89.2
FR	2.1	2.6	2.6	–	-0.5	-0.5	–	100.0	99.9	99.3	–
HR	2.0	2.7	2.9	3.1	-0.7	-0.9	-1.1	100.0	100.0	99.9	99.6
IE	4.0	5.0	5.2	5.5	-1.1	-1.2	-1.6	100.0	100.0	99.9	97.7
IT	5.8	7.7	7.9	–	-1.9	-2.1	–	100.0	100.0	99.3	–
LT	9.0	10.4	11.7	12.5	-1.4	-2.7	-3.5	100.0	100.0	99.8	98.6
LU	3.7	4.9	6.5	7.1	-1.2	-2.8	-3.3	100.0	100.0	99.9	96.3
LV	7.3	7.9	8.2	8.7	-0.6	-0.9	-1.4	100.0	97.9	87.3	82.6
MT	1.5	2.2	2.6	2.5	-0.7	-1.1	-1.1	100.0	100.0	98.8	95.0
NL	5.9	6.7	7.2	7.6	-0.8	-1.3	-1.7	100.0	99.9	99.7	87.8
SI	1.9	2.4	3.1	3.3	-0.4	-1.1	-1.3	100.0	99.8	99.5	99.0
SK	2.1	2.6	3.0	3.2	-0.5	-0.9	-1.1	100.0	100.0	99.9	92.4

Notes: The P20–P80 range is the survey-weighted 80th percentile minus the 20th percentile of household annual-average inflation, in percentage points. D2–D3, D2–D4 and D2–D5 report the coarser range minus the finer range. Household indices use fixed 2020 HBS baskets and annual expenditure-margin RAS calibration, except Malta at D5, where relative-expenditure weights are used because household RAS does not converge. Actual and imputed rents (HBS 041 and 042) and the corresponding HICP actual-rent component are excluded before aggregation. France, Italy and Spain use the dedicated digit-4 household files. Portugal is excluded because the required detailed household expenditures are unavailable. Coverage is the survey-weighted share of retained household expenditure matched to complete HICP price and weight series. A dash denotes unavailable detail.

C Robustness to Rent Treatment and Income Definition

C.1 Rent Treatment

The baseline removes actual and imputed rents before constructing matched product baskets and renormalizes the remaining weights. Table C.1 compares this baseline with two alternatives: mapping both actual and imputed rents to HICP actual rents, and retaining actual rents only. The sensitivity exercise uses relative-expenditure weights and therefore need not reproduce the RAS-based baseline estimates. It reports the individual P20–P80 range because rent treatment can alter inflation dispersion when rent shares differ across households.

C.2 Income Definition

The baseline ranks households by total net household income. Table C.2 tests whether this choice drives the measured inflation gap. We keep the household inflation rates, sample, consumption coverage, and weights fixed, and reassign national quintiles using equivalized income.

Table C.1: Robustness of individual P20–P80 inflation dispersion to rent treatment (2022)

Country	All rents	Actual rents only	Excluding rents
Austria	3.5	4.0	3.8
Belgium	6.5	7.1	7.9
Croatia	2.5	2.6	2.5
Cyprus	2.7	2.9	2.9
Estonia	10.8	10.8	11.1
Finland	3.4	4.6	3.7
France	3.0	3.0	2.6
Germany	4.0	4.0	4.2
Greece	4.2	4.9	4.6
Ireland	4.5	4.6	5.2
Italy	7.3	7.3	7.9
Latvia	7.4	7.8	7.7
Lithuania	9.9	10.1	10.2
Luxembourg	4.1	4.4	4.4
Malta	2.1	2.1	2.1
Netherlands	5.4	6.0	7.0
Slovakia	2.4	2.8	2.6
Slovenia	2.3	2.3	2.3
Spain	4.0	4.0	3.9
Euro area (19 countries)	4.5	4.6	4.8

Notes: Entries are the survey-weighted P80 minus P20 range of annual-average household inflation, in percentage points. All scenarios use fixed 2020 HBS baskets, relative-expenditure weights and chained Laspeyres indices at COICOP digit 3. All rents maps HBS actual and imputed rents (041 and 042) to HICP actual rents (041); Actual rents only retains HBS 041; Excluding rents removes HBS 041 and 042 and HICP 041 before normalization. The euro-area row averages national ranges with 2022 HICP country weights over the 19-country household sample; Portugal is excluded because the required detailed household expenditures are unavailable.

Table C.2: Robustness to income definition: inflation gap (in points, 2022)

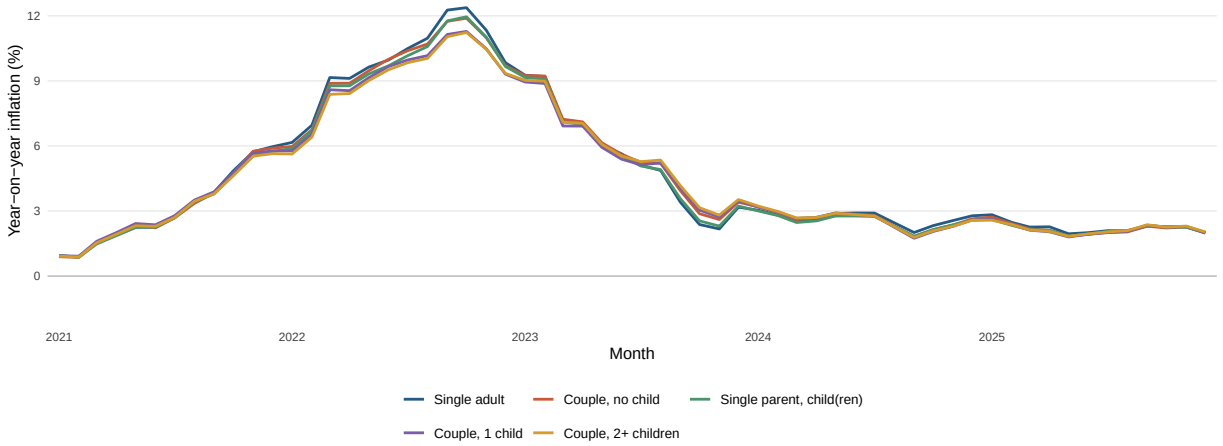
Country	Total household income	Equivalized income	Difference
Austria	0.2	0.2	0.1
Belgium	3.2	1.4	-1.8
Croatia	0.3	0.8	0.5
Cyprus	1.0	0.6	-0.4
Estonia	5.0	5.4	0.5
Finland	-0.6	-0.4	0.3
France	0.3	0.1	-0.2
Germany	1.8	1.7	0.0
Greece	1.4	0.7	-0.7
Ireland	4.0	2.1	-1.8
Latvia	5.2	4.0	-1.2
Lithuania	4.6	3.3	-1.4
Luxembourg	-0.1	0.0	0.1
Malta	0.2	0.0	-0.2
Netherlands	3.9	2.6	-1.3
Slovakia	0.1	0.2	0.1
Slovenia	0.6	0.4	-0.2
Spain	1.2	0.8	-0.4
Euro area (18 countries)	1.4	1.1	-0.3

Notes: The table reports first-quintile minus fifth-quintile annual-average household inflation in percentage points. Both columns use the same 2022 household inflation observations, the same rent-excluded consumption universe, RAS weights, survey weights, and country samples. Only the national quintile ranking changes. Total household income is EUR_HH099. Equivalized income divides EUR_HH099 by the OECD equivalence scale HB061. Difference is equivalized income minus total household income. The euro-area row averages national gaps with 2022 HICP country weights over the 18-country sample.

D Inflation by Household Composition

This appendix extends Figure 7 by grouping households according to family composition. The five categories are mutually exclusive: single adults; couples without children; single parents with one or more children; couples with one child; and couples with two or more children.

Figure D.1: Year-on-year inflation by household composition, 18-country HBS sample



Notes: Household-specific annual inflation rates are calculated from 2020 HBS expenditure baskets and monthly HICP prices at COICOP digit 3, using the same RAS and country-aggregation procedure as Figure 7. Family types are identified from MB05: a couple contains the reference person and a spouse or partner, and a child is a member recorded as the child of either adult. Households that also contain parents, other relatives, non-relatives, or unspecified members are excluded. Sources: Eurostat HBS 2020 scientific-use files and HICP; authors' calculations.

E RAS Weighting

We use the RAS (Row and Column Adjustment Scaling) algorithm. RAS is a bi-proportional matrix scaling method that alternately rescales rows and columns until the required marginal totals are matched (Bacharach, 1965). In our setting, we combine official HICP item weights with the relative HBS expenditure profile of each household group.

We distinguish three objects: the HICP product shares, the aggregate product-by-group expenditure mass calibrated by RAS, and the resulting within-group baskets. First, let $s_{g,y}$ denote the share of group g in total consumption expenditure used for HICP weight year y:

$$s_{g,y} = \frac{E^{\text{HBS}}_{g,y'}}{E^{\text{HBS}}_{\text{all},y'}}, \qquad \sum_g s_{g,y} = 1, \tag{13}$$

where y' is the HBS wave matched to HICP weight year y. In the baseline, $y' = 2020$ and the expenditure margins are computed from the same household sample as the group baskets. Historical applications select the most recent available preceding HBS wave, or the earliest wave when no preceding observation exists. These targets are expenditure shares, not population shares. For Italy, the income targets use the Istat-based HBS inputs described in Appendix G.

Let $\mathcal{J}^{\text{M}}_y$ be the set of matched COICOP categories available in the HICP weighting structure in year y. Let $E^{\text{HBS}}_{j,g,y'}$ denote expenditure on item j by group g in the matched HBS wave y', and let $E^{\text{HBS}}_{j,\text{all},y'}$ denote expenditure on that item by all households. Consistent with the implementation, the HBS correction factor used in the RAS seed is the share of aggregate expenditure on item j accounted for by group g:

$$c^{\text{HBS}}_{j,g,y'} = \frac{E^{\text{HBS}}_{j,g,y'}}{E^{\text{HBS}}_{j,\text{all},y'}}. \tag{14}$$

Thus $c^{\text{HBS}}_{j,g,y'}$ records how expenditure on item j is distributed across household groups. To avoid mixing index-weight units with expenditure shares, define the HICP weight normalized over the matched product set as

$$\bar{\omega}^{\text{HICP}}_{j,y} = \frac{\omega^{\text{HICP}}_{j,y}}{\sum_{k \in \mathcal{J}^{\text{M}}_y} \omega^{\text{HICP}}_{k,y}}, \qquad \sum_{j \in \mathcal{J}^{\text{M}}_y} \bar{\omega}^{\text{HICP}}_{j,y} = 1. \tag{15}$$

The initial product-by-group mass is then

$$Z_{j,g,y} = \max\left\{ s_{g,y} \bar{\omega}^{\text{HICP}}_{j,y} c^{\text{HBS}}_{j,g,y'}, \epsilon \right\}, \qquad \epsilon = 10^{-12}. \tag{16}$$

The small positive floor is a numerical safeguard to keep the RAS problem strictly positive after HICP-HBS matching. RAS transforms this seed into a joint expenditure-share matrix $M_{j,g,y}$.

Its group margins equal the groups' expenditure shares, while its product margins equal the normalized HICP shares:

$$\sum_{j \in \mathcal{J}_y^{\mathrm{M}}} M_{j,g,y} = s_{g,y} \quad \text{for all } g, \qquad \sum_{g} M_{j,g,y} = \bar{\omega}_{j,y}^{\mathrm{HICP}} \quad \text{for all } j. \tag{17}$$

Both sets of margins sum to one, so the constraints are mutually compatible.

The RAS algorithm starts from $M_{j,g,y}^{(0)} = Z_{j,g,y}$ and repeats two scaling steps. At iteration t, the group-margin adjustment is

$$\widetilde{M}_{j,g,y}^{(t)} = M_{j,g,y}^{(t-1)} \times \frac{s_{g,y}}{\sum_{\ell \in \mathcal{J}_y^{\mathrm{M}}} M_{\ell,g,y}^{(t-1)}}, \tag{18}$$

and the product-margin adjustment is

$$M_{j,g,y}^{(t)} = \widetilde{M}_{j,g,y}^{(t)} \times \frac{\bar{\omega}_{j,y}^{\mathrm{HICP}}}{\sum_{h} \widetilde{M}_{j,h,y}^{(t)}}. \tag{19}$$

We iterate these two steps by weight year until the maximum absolute row or column marginal error is below 10^{-10}, with a maximum of 1,000 iterations.

The calibrated matrix M is a joint expenditure distribution. The group-specific HICP basket is the corresponding conditional product distribution within group g:

$$\omega_{j,g,y} = \frac{M_{j,g,y}}{s_{g,y}} = \frac{M_{j,g,y}}{\sum_{k \in \mathcal{J}_y^{\mathrm{M}}} M_{k,g,y}}. \tag{20}$$

The equality of the two denominators follows directly from the group-margin constraint. It implies that each group basket sums to one. Moreover, the expenditure-share-weighted average of the group baskets reproduces the normalized HICP product weights:

$$\sum_{j \in \mathcal{J}_y^{\mathrm{M}}} \omega_{j,g,y} = 1, \qquad \sum_{g} s_{g,y} \omega_{j,g,y} = \bar{\omega}_{j,y}^{\mathrm{HICP}}. \tag{21}$$

When reported as percentages, the group-specific weights are $100\omega_{j,g,y}$ and therefore sum to 100 within every household group. This rescaling leaves the RAS constraints and the resulting price indices unchanged.

This preserves the relative information contained in the HBS expenditure profiles while making the aggregate of the household-group baskets consistent with the published HICP weighting structure, up to residual differences induced by product coverage and COICOP matching.

The all-items reference exercise in Table E.1 compares the two weights for Germany at COICOP digit 3 over the period beginning in 2019. The RAS calibration tracks the published German HICP

more closely than the relative-expenditure weighting, whose implicit aggregate weights are equal across groups.

Table E.1: Germany HICP validation by weighting scheme (differences in pp)

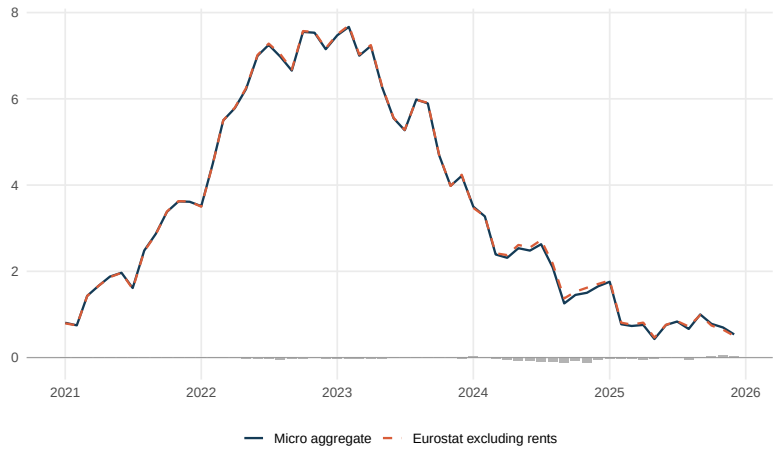
Method	Obs.	Mean diff.	Mean abs. diff.	RMSE	Max abs. diff.
RAS	88	-0.059	0.093	0.118	0.259
Relative expenditure	88	-0.070	0.143	0.166	0.449

Notes: Validation statistics compare the recalculated average of German income-quintile inflation rates with the published all-items HICP inflation rate. The sample is monthly, from January 2019 to April 2026. All differences are recalculated minus published inflation and are reported in percentage points.

F Validation Tests

Figure F.1 checks the internal aggregation against Eurostat on matching French non-rent coverage. Figure F.2 provides an external comparison with INSEE's published inflation by standard-of-living decile.

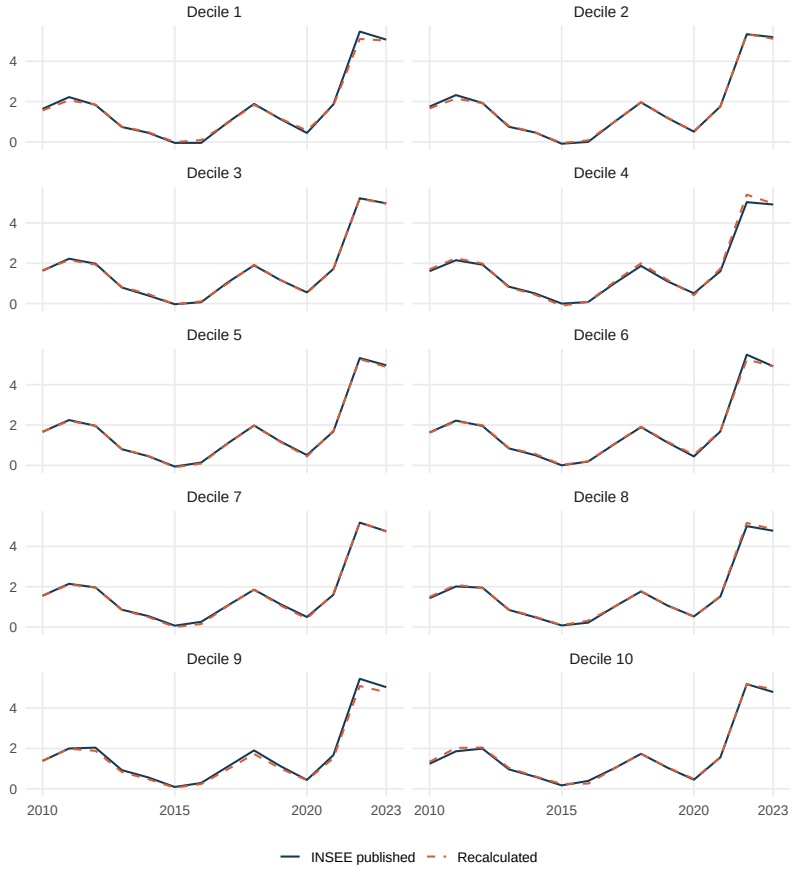
Figure F.1: France: micro aggregate and Eurostat excluding rents (year on year, in %)



Notes: Identical country, time, and non-rent coverage. Gray bars show the micro aggregate minus Eurostat, in percentage points.
Sources: HBS microdata, Eurostat HICP, authors' calculations.

The micro aggregate differs from the Eurostat benchmark by less than 0.1 percentage point on average. This agreement partly reflects RAS calibration and does not validate household heterogeneity or policy counterfactuals.

Figure F.2: External consistency: annual inflation by standard-of-living decile, France (in %)



Notes: Annual inflation, 2010–2023. The recalculation uses COICOP digit 4 and the original INSEE standard-of-living deciles.
Sources: INSEE, Eurostat, authors' calculations.

G Income-Group Baskets in Italy

This appendix explains the construction of Italy's income-group indices. The Italian files provide the three-digit expenditure vectors needed for household-level indices, so Italy enters the individual P20–P80 distributions without an income filter. They do not provide comparable income jointly with those household expenditure vectors. Italy therefore remains excluded from household-level saturated-cell regressions, while its income-group indices use the bridge below.

For income, we combine the 2005 harmonized income-quintile profile with the 2015 and 2020 Istat files. Households are ranked by Carbonaro-equivalized consumption using survey weights. Let E_{jqy} be expenditure on bridge group j by expenditure quintile q, $s^I_{jq,2005}$ and $s^E_{jq,2005}$ the 2005 income- and expenditure-quintile shares, and n_{jy} the national share. The script transfers the 2005 relationship and contracts the gradient toward the national margin:

$$\widetilde{s}^I_{jqy} = \frac{E_{jqy}\left(s^I_{jq,2005}/s^E_{jq,2005}\right)}{\sum_k E_{kqy}\left(s^I_{kq,2005}/s^E_{kq,2005}\right)}, \qquad \widehat{s}^I_{jqy} = \frac{n_{jy} + \lambda_j\left(\widetilde{s}^I_{jqy} - n_{jy}\right)}{\sum_k \left[n_{ky} + \lambda_k\left(\widetilde{s}^I_{kqy} - n_{ky}\right)\right]}. \tag{22}$$

Here λ_j is 0.20 for housing, 0.35 for transport and recreation, and 0.65 otherwise. The division targets are allocated to three-digit products using national within-division shares and RAS-calibrated to HICP weights. They are group-level bridge estimates, not household observations (Donatiello et al., 2022). Table G.1 reports the RAS group margins.

Table G.1: Income-group expenditure shares in Italy

Group	2015	2020
First quintile	7.8	8.1
Second quintile	12.7	12.8
Third quintile	17.2	17.0
Fourth quintile	23.3	22.8
Fifth quintile	39.0	39.3

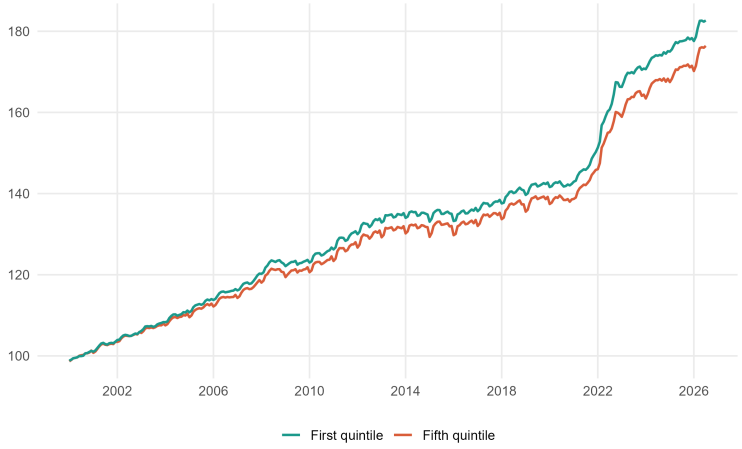
Notes. Shares of annual consumption used as RAS income-group margins. These are group-level inputs, not household records.

Sources: Istat HBS, Eurostat HBS, authors' calculations.

H Long-Run Inflation Inequality

Figure H.1 shows a modest but persistent divergence between the first- and fifth-quintile price indices after 2008. The post-COVID episode produces the largest visible widening in the series since 2000.

Figure H.1: Long-run inflation inequality in the euro area, by income quintile (index, 2000 = 100)



Notes: This figure shows monthly price indices for the first and fifth income quintiles in the euro area, January 2000 to July 2026. Indices are rebased to 2000 = 100.
Sources: Eurostat HICP, HBS, authors' calculations.

This pattern contrasts with the larger long-run gaps documented for the United States by Jaravel (2024). Several methodological differences may account for the contrasting results. First, the euro-area indices reported here aggregate national indices and therefore average across countries whose price dynamics and fiscal responses can offset one another. Second, the euro-area calculation uses harmonized COICOP categories, whereas the U.S. evidence exploits much finer product-level data. Aggregation within broad categories can conceal persistent differences in the prices paid by different income groups, which are captured in the U.S. data.

I Counterfactuals

This appendix reports the country-level policy inventory, sensitivity to alternative counterfactual constructions, the product-group decomposition of annual policy effects, and country-level observed and counterfactual headline inflation series. Each country figure combines all simulated energy and food price-policy layers using the HICP item weights in the national CPI basket.

I.1 Policy Inventory and Counterfactual Definitions

Tables I.1 and I.2 separate the coverage of the policy inventory from the rules used to construct counterfactual prices.

Table I.1: Price measures included in the national counterfactual estimates

Country	Simplified policy coverage	Policy years	Effect identification	Policy count
Austria	Products: electricity, gas. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2024	1 source-determined; 3 imputed (official formula, representative bill input).	4 entries; 3 distinct policies
Belgium	Products: electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2023	2 source-determined; 4 imputed.	6 entries; 3 distinct policies
Croatia	Products: food, electricity, gas, heating. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2023	7 source-determined; 2 imputed.	9 entries; 4 distinct policies
Cyprus	Products: electricity, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2021–2023	1 source-determined; 4 imputed.	5 entries; 3 distinct policies
Estonia	Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2021–2024	0 source-determined; 8 imputed.	8 entries; 8 distinct policies
Finland	Products: electricity, motor fuels, passenger transport. Instruments: tax and charge relief.	2022–2023	5 source-determined; 1 imputed.	6 entries; 3 distinct policies
France	Products: electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, subsidies or rebates.	2021–2024	2 source-determined; 1 imputed.	3 entries; 3 distinct policies
Germany	Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2024	1 source-determined; 6 imputed.	7 entries; 6 distinct policies
Greece	Products: electricity, gas, motor fuels. Instruments: subsidies or rebates.	2021–2023	0 source-determined; 3 imputed.	3 entries; 3 distinct policies
Ireland	Products: electricity, gas, motor fuels. Instruments: tax and charge relief, subsidies or rebates.	2022–2023	2 source-determined; 5 imputed.	7 entries; 7 distinct policies

Continued on next page

Country	Simplified policy coverage	Policy years	Effect identification	Policy count
Italy	Products: electricity, gas, motor fuels. Instruments: tax and charge relief, subsidies or rebates.	2021–2023	3 source-determined; 5 imputed.	8 entries; 5 distinct policies
Latvia	Products: electricity, gas, heating. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2023	0 source-determined; 3 imputed.	3 entries; 3 distinct policies
Lithuania	Products: electricity, gas. Instruments: subsidies or rebates.	2022–2023	0 source-determined; 2 imputed.	2 entries; 1 distinct policy
Luxembourg	Products: electricity, gas, heating, motor fuels, other consumption. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2022–2024	67 source-determined; 5 imputed.	72 entries; 6 distinct policies
Malta	Products: electricity, motor fuels. Instruments: price caps or regulated tariffs, subsidies or rebates.	2022–2023	0 source-determined; 2 imputed.	2 entries; 2 distinct policies
Netherlands	Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief.	2022–2023	7 source-determined; 4 imputed.	11 entries; 6 distinct policies
Portugal	Products: food, electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief.	2021–2026	1 source-determined; 14 imputed.	15 entries; 8 distinct policies
Slovakia	Products: electricity, gas, heating. Instruments: price caps or regulated tariffs, tax and charge relief.	2023	0 source-determined; 3 imputed.	3 entries; 1 distinct policy
Slovenia	Products: electricity, gas, heating, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief.	2021–2023	0 source-determined; 9 imputed.	9 entries; 7 distinct policies
Spain	Products: food, electricity, gas, motor fuels. Instruments: price caps or regulated tariffs, tax and charge relief, subsidies or rebates.	2021–2026	5 source-determined; 10 imputed.	15 entries; 9 distinct policies
Euro area	**Sum of the 20 country rows.**	**2021–2026**	**104 source-determined; 94 imputed.**	**198 entries; 91 distinct policies**

Notes: Products and instruments are grouped for readability; the underlying workbook retains the measure-level descriptions, dates, COICOP codes, and the measure-specific evidence basis. Source-determined effects are quantified directly from statutory tax rates, regulated tariffs, or official administrative price or bill series. Imputed effects require an aggregate-equivalent wedge, representative bill, calibrated pass-through, or model-based proxy. An official policy announcement alone does not make an effect source-determined when its price effect must be estimated. Policy count reports executed country-product entries and distinct policy devices. The distinct count consolidates successive phases or extensions and a single legal measure represented by several product trajectories; for example, Luxembourg's product-specific VAT trajectories count as one policy. The Euro-area row is the sum of the 20 country rows. Sources: Bruegel Dataset (2021), 'National policies to shield consumers from rising energy prices,' version of 26 June 2023; authors' calculations.

Table I.2: Official energy-price-support rules and their counterfactual implementation

Country and policy	Official rule	Implementation	Main difference
Austria: electricity price brake	$S = \min(Q, 2{,}900)\min\{\max(p - 0.10, 0), 0.30\}$; $B^1 = B^0 - S$.	E-Control bill: $Q = 3{,}500$, $p =$ EUR 0.24/kWh; official non-energy components and VAT. $\widehat{\rho} = B^0/B^1 = 1.433$.	Statutory rule on a representative bill; no contract-price, consumption, or billing-date distribution.
Germany: electricity and gas price brakes	Relief: $S = \alpha Q^0 \max(p - c, 0)$; bill: $B^1 = F + pQ - S$. $\alpha = 0.8$; caps: EUR 0.40/kWh (electricity), EUR 0.12/kWh (gas).	Same bill rule; representative contract. p: BNetzA April 2023 price, monthly HICP path. Baseline: $Q/Q^0 = 1$, $F = 0$; $\widehat{\rho} = B^0/B^1$.	Contract-level rule versus representative contract; no distribution of prices, fixed charges, or consumption.
Netherlands: 2023 electricity and gas price cap	Relief: $S = \min(Q, \bar{Q})\max(p - c, 0)$. Caps: EUR 0.40/kWh up to 2,900 kWh; EUR 1.45/m^3 up to 1,200 m^3.	CBS monthly contract and capped tariffs. $B^0 = B^1 + Q(p - p^{\text{cap}})$; $\widehat{\rho} = B^0/B^1$. Representative quantities: 2,063 kWh and 1,136 m^3.	Average bills and quantities versus household billing dates, consumption, and contract tariffs.
Italy: electricity and gas social bonuses	Bill credit: $S_{iq}(d) = CCI_{iq}d/D_q$, plus ordinary bonus. CCI_{iq}: quarter- and beneficiary-specific.	ARERA aggregate funding: $\widehat{\rho}_{jq} = E^0_{jq}/(E^0_{jq} - S_{jq})$; E^0_{jq}: domestic volume $\times$ average final price.	Means-tested bill credit versus aggregate HICP wedge; no eligibility, family-size, heating-use, or take-up incidence.
Spain: Iberian electricity-market adjustment	Generator adjustment: $Y_d = \max\{(P^{\text{gas}}_d - P^{\text{ref}}_d)/0.55, 0\}$ per MWh; cost allocated to non-exempt demand.	OMIE hourly settlements: $P^1_h = P^{\text{OMIE}}_h + A^C_h$; $P^0_h = P^{\text{OMIE}}_h + A^P_h$; $\rho_h = P^0_h/P^1_h$.	Published wholesale wedge; no recomputation of Y_d or retail exemptions; proportional HICP transmission.
France: gas and electricity tariff shields	No reconstructed formula: CRE and Insee publish observed and no-shield regulated-tariff paths.	Published tariff-path ratio applied to HICP; documented Insee pass-through parameters where required.	Public counterfactual tariff; only HICP mapping requires chaining and pass-through assumptions.

Notes: F: fixed charge; p: contract price; Q: current consumption; Q^0: reference consumption; c: statutory cap; A^C and A^P: OMIE consumer and producer adjustments. $\widehat{\rho}$ denotes a counterfactual-to-observed ratio reconstructed from representative or aggregate inputs; ρ_h is the corresponding directly observed hourly ratio. The table contrasts official rules with the HICP counterfactual transformation. Sources: Austrian Parliament and E-Control; Bundesnetzagentur and BMWK; Dutch Government and CBS; ARERA; Spain's Royal Decree-law 10/2022 and OMIE; CRE and Insee; authors' calculations.

I.2 Sensitivity to Counterfactual Construction

Table I.3 compares the full specification with direct reconstructions only, the exclusion of conditional benchmarks, and alternative benchmark wedges.

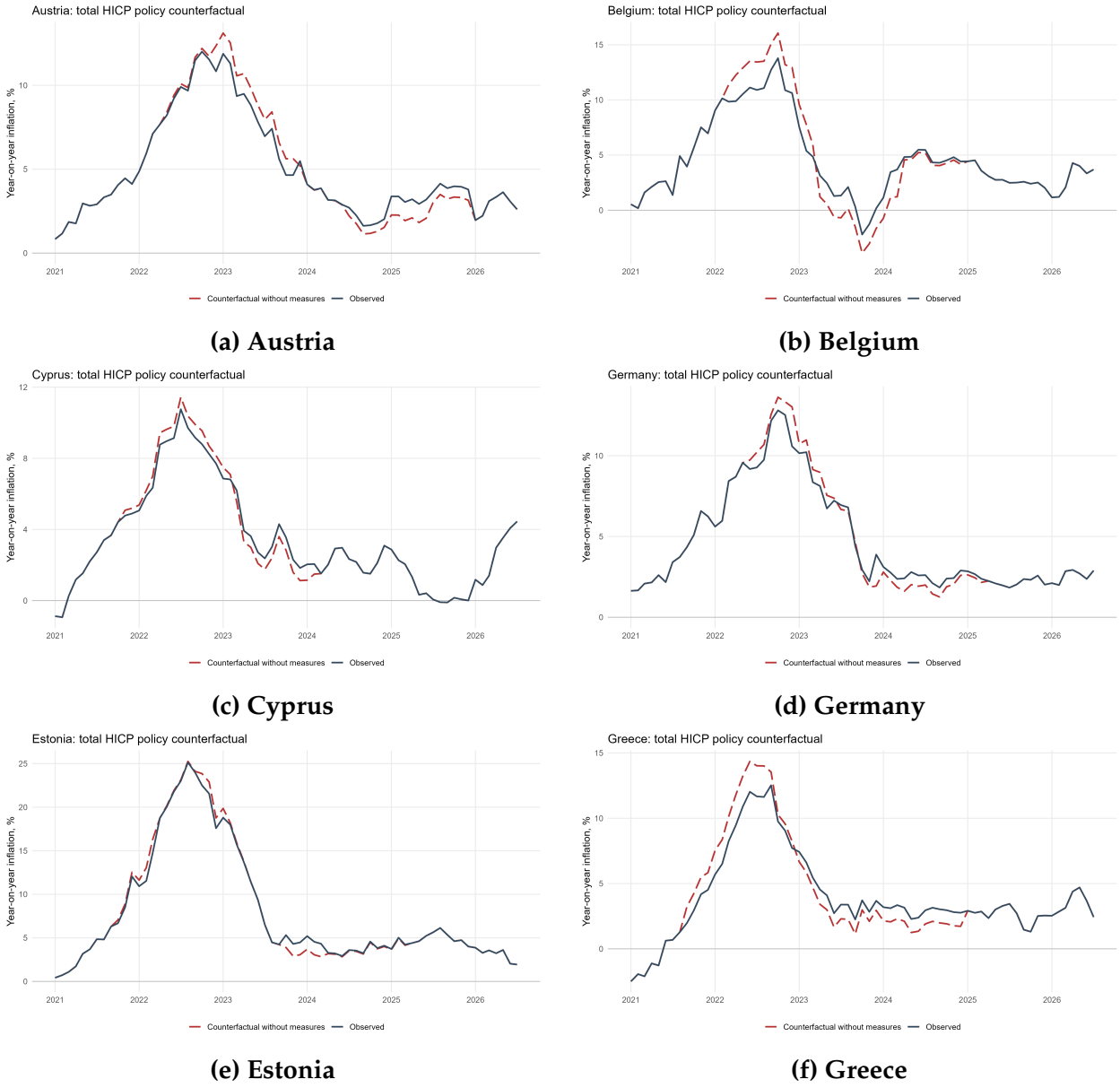
Table I.3: Sensitivity of euro-area inflation-inequality effects to counterfactual construction methods

	Inflation inequality					
	Individual P20–P80			Income Q1–Q5		
Scenario	2021	2022	2023	2021	2022	2023
Full specification	-0.1	-1.8	-0.7	-0.1	-0.6	-0.6
Direct reconstructions only	0.0	-1.1	-0.2	0.0	-0.4	-0.1
Full specification without conditional benchmarks	-0.1	-1.7	-0.4	-0.1	-0.6	-0.3
Conditional benchmark wedges at 50%	-0.1	-1.7	-0.5	-0.1	-0.6	-0.4
Conditional benchmark wedges at 150%	-0.1	-1.8	-0.9	-0.1	-0.7	-0.7
Dependence on estimated layers: full minus direct-only	-0.1	-0.7	-0.4	0.0	-0.3	-0.4

Notes. The first block reports the observed minus counterfactual P20–P80 range of annual household inflation, in percentage points. The range is P80 minus P20. National percentile endpoints use HBS survey weights and are averaged with annual HICP country weights over the 19-country household sample; Portugal is excluded. The second block reports the corresponding Q1–Q5 effects for all 20 euro-area countries. Direct reconstructions use statutory rates, unit rebates, or published reference-price paths without a fitted intensity. The 50% and 150% rows scale only the conditional benchmark price wedges. Full minus direct-only measures dependence on estimated layers; it is not a confidence interval. Sources: Eurostat, HBS 2020 microdata, national sources, authors' calculations.

I.3 Country-Level Observed and Counterfactual Inflation

Figures I.1–I.4 compare observed inflation with the counterfactual that removes household price policies.

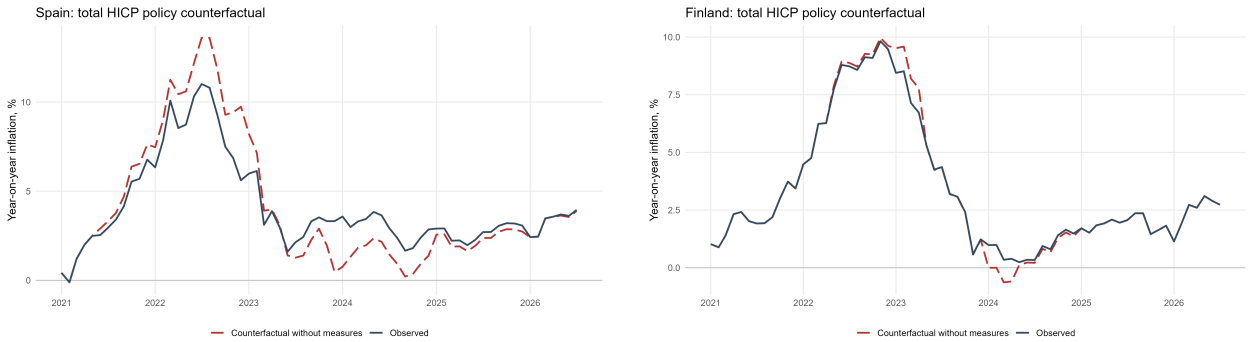


(a) Austria **(b) Belgium**

(c) Cyprus **(d) Germany**

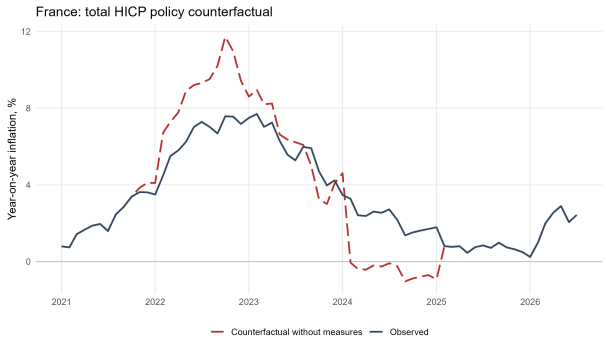
(e) Estonia **(f) Greece**

Figure I.1: Observed and counterfactual inflation: Austria, Belgium, Cyprus, Germany, Estonia, Greece

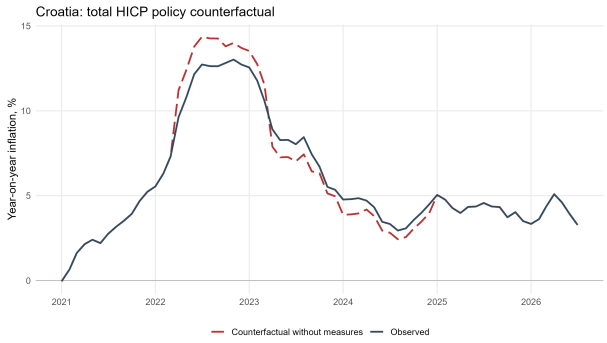


(a) Spain

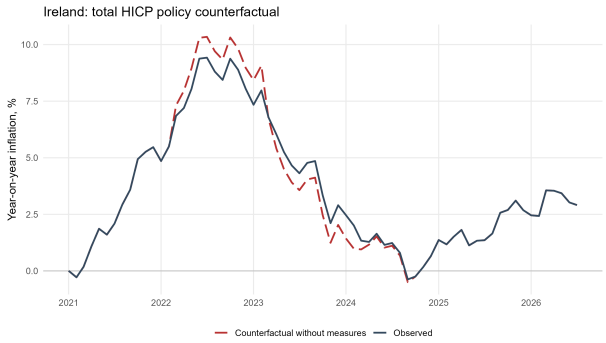
(b) Finland



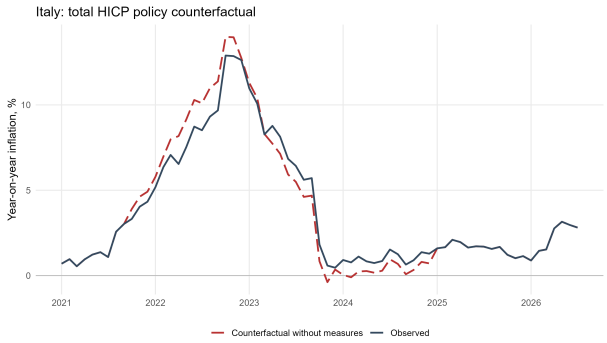
(c) France



(d) Croatia

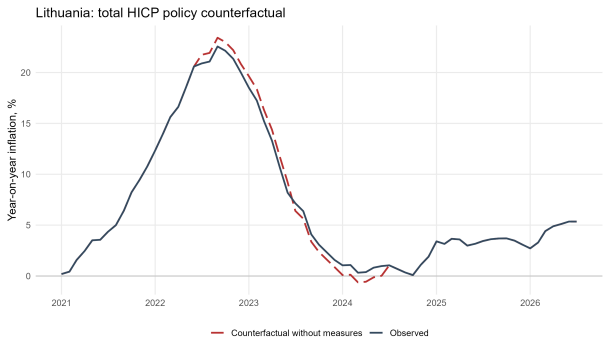


(e) Ireland

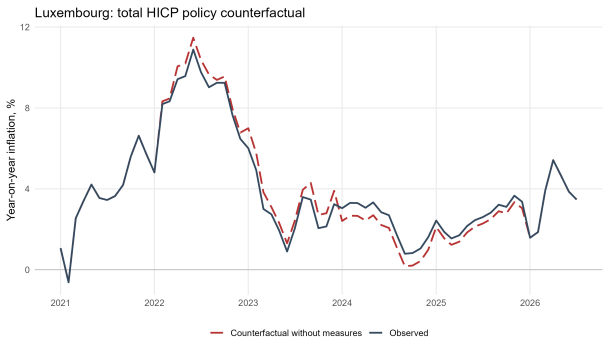


(f) Italy

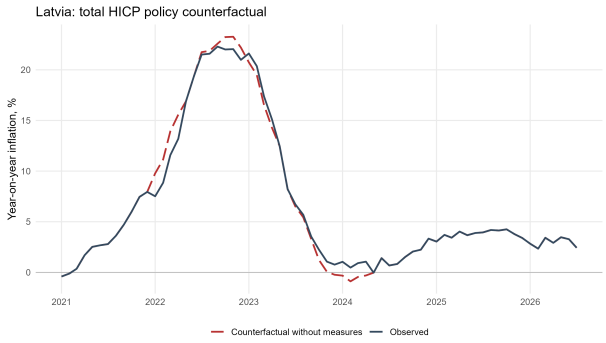
Figure I.2: Observed and counterfactual inflation: Spain, Finland, France, Croatia, Ireland, Italy



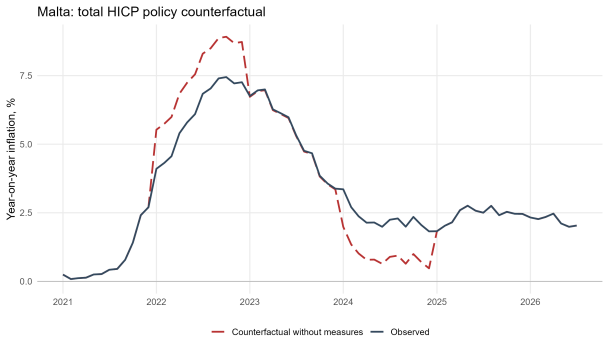
(a) Lithuania



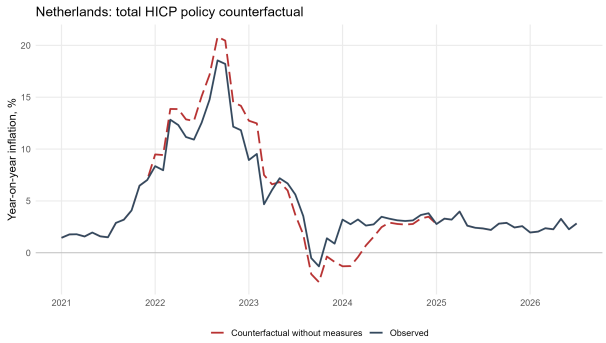
(b) Luxembourg



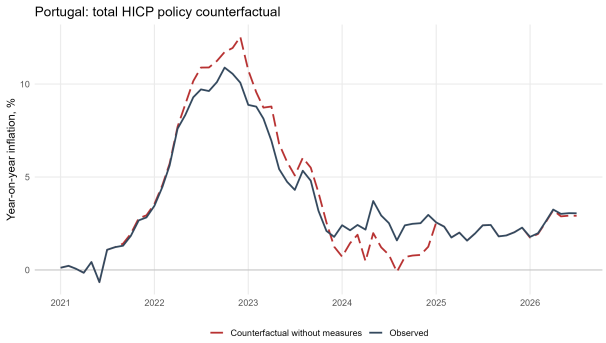
(c) Latvia



(d) Malta

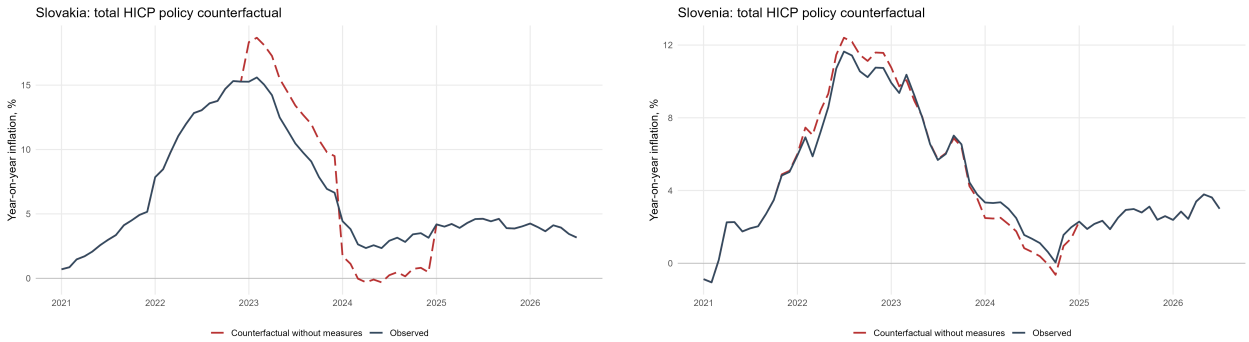


(e) Netherlands



(f) Portugal

Figure I.3: Observed and counterfactual inflation: Lithuania, Luxembourg, Latvia, Malta, Netherlands, Portugal



(a) Slovakia **(b) Slovenia**

Figure I.4: Observed and counterfactual inflation: Slovakia, Slovenia

I.4 Product Decomposition of Policy Effects

Table I.4 decomposes the aggregate effects by product group. Housing energy accounts for most of the reduction, while vehicle-operation expenditure partly offsets the fall in individual dispersion.

Table I.4: Product-group contributions of household price policies to individual P20–P80 inflation dispersion and Q1–Q5 inflation inequality, 2021–2023

Product group	Weight (2022, %)	Inflation inequality					
		Individual P20–P80			Income Q1–Q5		
		2021	2022	2023	2021	2022	2023
Food and non-alcoholic beverages	17.8	-0.2	-0.3	0.3	0.0	0.0	0.0
Alcoholic beverages, tobacco and narcotics	4.6	0.0	0.2	-0.4	0.0	0.0	0.0
Clothing and footwear	5.6	-0.1	-0.1	0.0	0.0	0.0	0.0
Housing maintenance and furnishings	8.0	0.1	0.3	0.5	0.0	0.0	0.0
Water, electricity, gas and other fuels	6.8	-0.8	-4.3	-1.1	-0.1	-0.7	-0.5
Operation of personal transport equipment	6.3	0.0	2.0	-0.1	0.0	0.1	0.0
Transport (excluding vehicle operation)	9.6	0.3	-1.2	-0.2	0.0	0.0	0.0
Other	41.1	0.6	1.6	0.3	0.0	0.0	0.0
Total	**100.0**	**-0.1**	**-1.8**	**-0.7**	**-0.1**	**-0.6**	**-0.6**

Notes. Policy effects are observed minus counterfactual product contributions, in percentage points. The first block decomposes the effect on the P80 minus P20 range of annual household inflation, evaluating each scenario at its own national percentile households and averaging national contributions with annual HICP country weights over the 19-country household sample; Portugal is excluded. The second block decomposes the Q1 minus Q5 gap for all 20 euro-area countries. Annual rates compare annual-average indices. Product weights are 2022 EA20 expenditure shares. Sources: Eurostat, HBS 2020 microdata, national statistical institutes, authors' calculations.